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Dispersive Coupling of Superconducting
Qubits

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Preface

Cavity quantum electrodynamics (CQED) has provided beautiful experiments in studies of light and matter interactions in the last decades. For the investigation of the Cavity QED, Jaynes-Cummings model(JCM) is usually invoked, which relies on the rotating-wave approximation (RWA) that works well for coupling strength g in optical regime.

Recently, a new domain, called “Circuit quantum electrodynamics” has started to gain interest, in which the role of atom and cavity are now played by the superconducting circuits. Due to the flexibility of the design and strong coupling strength, circuit QED can in turn explore new regimes. One that has attracted attention very recently is the ultrastrong coupling regime, which lead to breakdown of the well-established Jaynes-Cummings model and has been realized experimentally in superconducting circuits.

Intrigued by the accomplishment of the superconducting circuits in unraveling this rather unexplored physics beyond RWA, we set path to investigate the ultrastrong coupling regime and superconducting qubits, which is one of the promising candidates for constructing qubits for quantum computer.

I would like to express my gratitude to Assoc. Prof. Valerio Scarani for his dedicated supervisory, insightful discussions and valuable advices. It was such a great experience and opportunity for me to learn and develop my skills throughout the research. I would like to thank Jiri Minar and Wang YiMin for providing innovative ideas and extensive assistance in terms of the derivations of theoretical expressions and understanding of im-

portant elements in the subjects, without whom the completion of this thesis would also not have been possible.

On personal notes, I wish to thank my dear family and fellow friends for their continuing understanding, patience and support. Last but not least, I would like to thank the members of Valerio's group, Conneqt at Centre of Quantum Technologies for valuable feedback and advices, not to forget the laughter and tears we had together.

Abstract

In this research project, we studied and thereby, attempt to elucidate the coupling between qubits and resonator in both of the strong and ultrastrong regime. To facilitate a better understanding of the frequent deployed Rotating Wave Approximation (RWA) in Jaynes-Cummings model, we first reviewed the time averaging method and show that increase in coupling strength indeed leads to breakdown of RWA.

We then reformulate the dispersive theory of qubit oscillator dynamics by taking into account the trace of couplings associated with the co-rotating terms and counter-rotating terms. A general form of dispersive Hamiltonian for both one qubit and multi-qubits scenario are derived, and we show that our general expression indeed give rise to the correct forms in RWA and Non-RWA (NRWA) regime. Under appropriate constraints and conditions, the behavior of the dispersive shift and interqubit coupling strength in both regimes are compared and discussed. We further investigate the transition between RWA to NRWA by constructing an approximation called Intermediate RWA (IRWA) based on our general expression and time averaging method.

Lastly, we proposed a scheme to realize the dispersive qubit-resonator dynamics which involves a hybrid quantum circuit consists of different types superconducting qubits coupled via a mechanical harmonic oscillator known as nanomechanical resonator. This switchable and controllable coupling can be utilized for quantum information processing such as quantum gate operation and entanglement generation.

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Chapter 1

Introduction

More than 40 years ago, Jaynes and Cummings introduced a fully quantum mechanical model for the interaction of light and matter, which describes a wide range of physical phenomena and serves as the “standard model” of quantum optics, in particular the field of cavity quantum electrodynamics (CQED), where the atom interacts with single mode of the field inside the cavity.

In the strong coupling regime, when the coupling strength g between atom and the cavity exceeds the dissipation rate of both atom and cavity, κ and γ , the system exhibits coherent light matter oscillation and superposition states. Although the Hamiltonian of the atom field system contains the so-called counterrotating terms, for small normalized coupling strength with respect to cavity mode frequency(g/ω), they can be safely neglected by invoking rotating wave approximation(RWA), which give rise to the *Jaynes-Cummings model*.

In the past decade, there has been tremendous progress in the experimental development of a quantum computer, defined as a machine that could exploit quantum mechanical phenomena such as superposition and entanglement of the quantum bits (qubits) to solve computational problem[12]. Quantum computer(QC) should be understood as something analogous to laser, which serves for different purposes compared to the conventional light sources. One of the motivations for the quantum computer is the famous Shor’s quan-

tum algorithm, which allows factorization of large numbers to be done much more faster compared to classical supercomputers.¹ In 1982, Feynman also conjectured that quantum computers would be able to simulate quantum mechanical systems with a much greater degree of accuracy than is possible with classical computers.

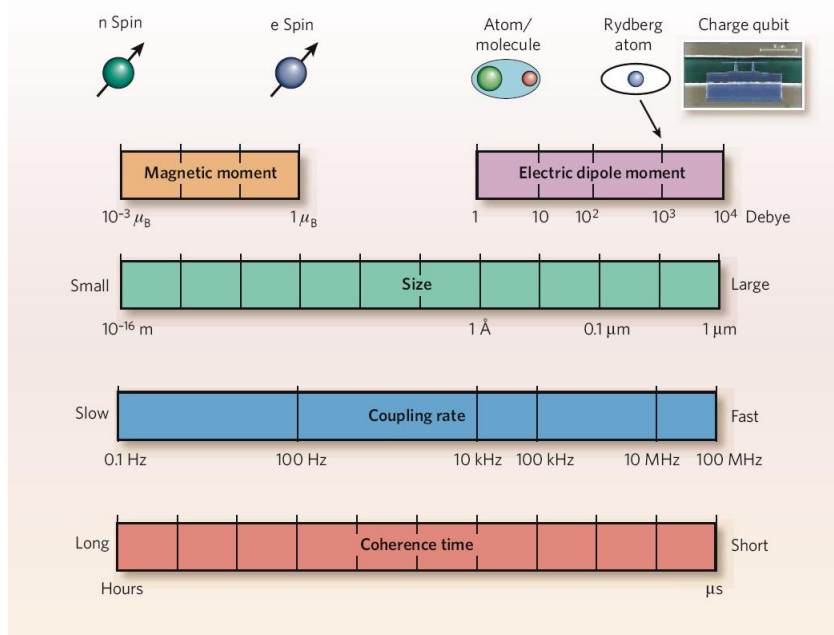


Figure 1.1: Comparison between the scalability, coupling strength and coherence time between different types of quantum system. Adapted from [19]

The main challenges for quantum computers are scalability, coupling strength and decoherence. Apart from that, the candidates also need to satisfy certain criteria known as “DiVincenzo” criteria, which can be summarized as: qubit existence, reset, coherence, gate operation and read out. Various types of physical system that satisfies these has been investigated for their potential as qubits, which can be divided into two classes in general[19].

The first class comprises “natural” candidates, such as single atoms, ions and nuclear spins[12]. They are well characterized and have long coherence time, but to scale the system up to a large number of qubit is not trivial. The second class of qubits is formed

¹This imposed a treat to the security of RSA encryption which is based on difficulty of factoring large number! However the number of qubits required is much larger anything demonstrated to date so far.

by “artificial” quantum system, which is typically in the form of solid state. Examples are quantum dots and superconducting circuits. They have the advantage of flexibility in design, good in scalability and easy initialization. The solid state system offers stronger coupling compared to the “natural” candidates, but as a result they are more prone to the decoherence from the environment. A comparison with scales of various properties is shown in Fig.1.1. In terms of coupling strength, charge qubit (one form of superconducting qubit) is of the order of 100MHz, but having the shortest coherence time, which is only about $1\mu s$.

One that has gained interest is the ultrastrong coupling regime, which is beyond the RWA and has been experimentally demonstrated in superconducting qubits [17, 8]. Among the consequences resulting from ultrastrong coupling, we are in particular interested in the dispersive regime, in which the qubit and resonator are far detuned with each other. Hence, this thesis will be devoted to the studies of dispersive regime of qubit-resonator dynamics in both RWA and non-RWA, followed by a proposal of controllable coupling between superconducting qubits in dispersive regime as demonstration.

The thesis is organized as follows. We begin with a brief review on atom field interaction (Chapter 2), in which Cavity Quantum Electrodynamics (CQED) and the Jaynes Cummings model are introduced. To gain more insight on the rotating wave approximation (RWA), time averaging method was studied in detail. We then briefly discussed some of the consequences of ultrastrong coupling, such as Bloch-Siegert Shift and recent experimental realization of ultrastrong coupling regime in superconducting circuit. Dispersive qubit-resonator dynamics is studied next, where we discuss the constraints and conditions in the dispersive regime and derive a general expression that applies to both RWA and Non-RWA (NRWA). By taking into account of the constraints, the properties of the dispersive shift and interqubits coupling of both RWA and NRWA is investigated. Next some background on the superconducting circuits is reviewed, in particular the charge and flux qubit. Finally we proposed a scheme to realize the dispersive regime in a hybrid

quantum circuit system which is coupled via a mechanical resonator.

Statement of Originality

My original works are mainly presented in Chapter 3 and 5, which includes:

- Derivation for the general form of the dispersive Hamiltonian in term of time-averaging function.
- Plots for resonator frequency shift and interqubit coupling strength for both RWA and NRWA case under the constraints and conditions on the dispersive Hamiltonian.
- Intermediate RWA from time-averaging function that indicates the transition from RWA to NRWA.
- Coupling of charge and flux qubit via nanomechanical resonator.

Chapter 2 and 4 serves to highlight the key concepts, ideas and background which I felt were necessary for understanding of the project. While they are not original, they have been summarized from my studies of various sources and presented in a way which I hope would be most coherent and illuminating.

Chapter 2

Atom-Photon interaction

In this chapter, we briefly discuss the interaction of the quantized radiation of field with the two-level atomic system described by Hamiltonian in the rotating wave approximation(RWA), in which in a single mode field it reduces to Jaynes-Cummings Model. To gain more insight on the RWA, time averaging method is studied in detail. We then briefly discuss some of the consequences arises from ultrastrong coupling and their recent experimental realization in superconducting circuit.

2.1 Interaction Hamiltonian

2.1.1 Two-Level Atom

In the dipole approximation, the atom -field interaction is analogous to a spin-1/2 particle in an external magnetic field. The interaction of a radiation field $\mathbf{E}$ with an atom can be described by the following Hamiltonian:

$$H = H_A + H_F + H_{\text{int}} \quad (2.1.1)$$

where H_A and H_F corresponds to the energy of the atom and the radiation field respectively. H_{int} is the interaction energy between the atom and field, in which in the case of electromagnetic field this will be the potential energy of the electric dipole interaction.

Two level approximation

Real atoms have complicated energy level structures, however, in many instances, only two atomic energy levels play a significant role in the interaction with the electromagnetic field. Under resonance condition, energy levels that lie far from the resonance can be ignored. In addition, selection rules imply that only certain energy levels are allowed for atomic transition. Hence the atomic Hamiltonian H_A can be written in the form:

$$H_A = E_e|e\rangle\langle e| + E_g|g\rangle\langle g| \quad (2.1.2)$$

$$= \frac{1}{2}(E_e + E_g)\hat{I} + \frac{1}{2}(E_e - E_g)\sigma_z \quad (2.1.3)$$

where $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$ and $\hat{I}$ is the identity operator. By redefining the atomic energies to be in the middle of the atomic transition, we then write

$$H_A = \frac{1}{2}\hbar\omega_0\sigma_z, \quad \omega_0 = \frac{(E_e - E_g)}{\hbar} \quad (2.1.4)$$

The free field Hamiltonian, after dropping the zero-point energy term, can be given in terms of the creation a and annihilation operators $a^\dagger$ ¹

$$H_F = \sum_{\mathbf{k}} \hbar\omega_k \left(a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} \right) \quad (2.1.5)$$

where ω_k is the frequency corresponds to the mode $\mathbf{k}$. Under dipole approximation, where the electric field is assumed to be uniform over the extent of the atom, the interaction Hamiltonian is

$$H_{\text{int}} = -\hat{\mathbf{d}} \cdot \hat{\mathbf{E}} \quad (2.1.6)$$

where

$$\hat{\mathbf{d}} = e\hat{\mathbf{r}} = \sum_{ij} e|i\rangle\langle i|\hat{\mathbf{r}}|j\rangle\langle j| \quad (2.1.7)$$

$$= \mathbf{d}\sigma_- + \mathbf{d}^*\sigma_+ = \mathbf{d}(\sigma_+ + \sigma_-) \quad (2.1.8)$$

¹In single mode, the operators satisfies $[a, a^\dagger] = 1$, with $a|n\rangle = \sqrt{n}|n-1\rangle$, $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$

in which by parity conservation $\langle i|\hat{\mathbf{r}}|i\rangle = 0$. We set $e\langle e|\hat{\mathbf{r}}|g\rangle = \mathbf{d}$ and introduce the atomic raising and lowering operators $\sigma_+ = |e\rangle\langle g|$ and $\sigma_- = |g\rangle\langle e|$. Without loss of generality we assumed d is real as well. The electric field represented by a collection of quantized harmonic oscillator takes the form

$$\mathbf{E} = \sum_{\mathbf{k}} \mathcal{E}_{\mathbf{k}} \left(a_{\mathbf{k}} + a_{\mathbf{k}}^\dagger \right) \quad (2.1.9)$$

where $\mathcal{E}_{\mathbf{k}} = (\hbar\omega_k/2\epsilon_0 V)^{1/2} \mathbf{e}_{\mathbf{k}}$. Thus the interaction Hamiltonian is

$$H_{\text{int}} = \hbar \sum_{\mathbf{k}} g_{\mathbf{k}} (\sigma_+ + \sigma_-) \left(a_{\mathbf{k}} + a_{\mathbf{k}}^\dagger \right) \quad (2.1.10)$$

with $g_k = -\mathbf{d} \cdot \mathcal{E}_{\mathbf{k}}/\hbar$, which is known as coupling strength. On substituting the terms above, the total Hamiltonian 2.1.1 can be expressed as

$$H = \frac{1}{2} \hbar \omega_0 \sigma_z + \sum_{\mathbf{k}} \hbar \omega_k a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \hbar \sum_{\mathbf{k}} g_{\mathbf{k}} (\sigma_+ + \sigma_-) \left(a_{\mathbf{k}} + a_{\mathbf{k}}^\dagger \right) \quad (2.1.11)$$

2.2 Rotating Wave Approximation

2.2.1 Interaction Picture

In the interaction picture (See A.1), the total Hamiltonian become²

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle_I = H_{\text{int}} |\psi\rangle_I \quad (2.2.1)$$

which implies the changes of state in the interaction picture are proportional to g the coupling strength. Under weak field condition[5], where $g \ll \omega_0$, the state does not evolve much over an optical period when ω_0 , the atomic frequency is in optical regime. This serves as our basis of the non-perturbative schemes called *Rotating Wave Approximation*.

²For simplicity the superscript I for interaction picture on operator is omitted

The full Hamiltonian Eq. (2.1.11) in the Schrödinger picture can be expressed as

$$H = H_0 + H_{\text{int}} \quad (2.2.2)$$

with $H_0 = H_A + H_F$. In the interaction picture, the operators are transformed accordingly by either Eq. (A.1.4) or Eq. (A.1.6), giving

$$a_{\mathbf{k}}(t) = a_{\mathbf{k}} e^{i\omega_k t}, \quad \sigma_z(t) = \sigma_z, \quad \sigma_{\pm}(t) = e^{\pm i\omega_o t} \sigma_{\pm} \quad (2.2.3)$$

Substituting these into H_{int} and splitting the term into two contribution called co-rotating term $H_{\text{int}}^{(r)}$ and counter-rotating term $H_{\text{int}}^{(ar)}$, we have

$$H_{\text{int}}(t) = H_{\text{int}}^{(r)}(t) + H_{\text{int}}^{(ar)}(t) \quad (2.2.4)$$

where

$$H_{\text{int}}^{(r)}(t) = \hbar \sum_{\mathbf{k}} g_{\mathbf{k}} \left(a_{\mathbf{k}} \sigma_+ e^{i(\omega_o - \omega_k)t} + a_{\mathbf{k}}^\dagger \sigma_- e^{-i(\omega_o - \omega_k)t} \right) \quad (2.2.5)$$

and

$$H_{\text{int}}^{(ar)}(t) = \hbar \sum_{\mathbf{k}} g_{\mathbf{k}} \left(a_{\mathbf{k}} \sigma_- e^{-i(\omega_o + \omega_k)t} + a_{\mathbf{k}}^\dagger \sigma_+ e^{i(\omega_o + \omega_k)t} \right) \quad (2.2.6)$$

However, this expression cannot be diagonalized exactly, and is often simplified by the rotating wave approximation (RWA). The standard arguments are:

Violation of Energy Conservation

A heuristic argument can be done by claiming that the term $\sigma_+ a^\dagger$ and $\sigma_- a$ do not conserve the energy, in which the former corresponds to excitation of atom during the emission of a photon, and the latter term describes the process of absorbing a photon and dropping to a lower energy level. On the contrary, the terms in $H_{\text{int}}^{(r)}$ only have either excitation/de-excitation of atom or field in the same time. As a result these non-energy conserving terms in $H_{\text{int}}^{(ar)}$ are discarded. [20].

Method of averaging

We noticed that the co-rotating term and counter-rotating contain oscillatory terms which involved the sum and difference of the field and atomic transition respectively. For near resonant case, the difference term, $\Delta = \omega_0 - \omega$ are much smaller than the sum $\omega_0 + \omega$. Integration of the time-dependent Schrödinger equation will bring the frequency sum and difference into the denominator. The fast oscillation term will hence be averaged out over any meaningful timescale, and the slow varying part dominates.

Preserving the co-rotating term, the resulting interaction Hamiltonian back in the original Schrödinger picture is thus

$$H_{\text{int}}(t) = \hbar \sum_{\mathbf{k}} g_{\mathbf{k}} \left(a_{\mathbf{k}} \sigma_+ + a_{\mathbf{k}}^\dagger \sigma_- \right) \quad (2.2.7)$$

2.2.2 Time averaging method

In order to gain better understanding on RWA, we resort to the *time averaging method*, in which the time scale are separated explicitly by means of temporal filtering. Following the [5], we introduce the averaging function $\varpi(t)$ that satisfied the properties below:

(1) Positive and even

$$\varpi(t) = \varpi(-t) \quad (2.2.8)$$

(2) Normalized

$$\int_{-\infty}^{\infty} dt \varpi(t) = 1 \quad (2.2.9)$$

(3) Peaked at $t=0$

Functions such as delta function and Gaussian function satisfy the conditions above. A simple example of such function is

$$\varpi(t) = \begin{cases} \frac{1}{\Delta T} & \text{for } -\frac{\Delta T}{2} \leq t \leq \frac{\Delta T}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2.2.10)$$

The average of a function is defined by the convolution

$$\overline{f(t)} = \int_{-\infty}^{\infty} dt' \varpi(t-t') f(t') = \int_{-\infty}^{\infty} dt' \varpi(t') f(t+t') \quad (2.2.11)$$

where ϖ has a temporal width ΔT that washes out oscillation with period smaller than ΔT . For our purpose, it is more convenient to work in the domain of frequency, by using the Fourier transformed time averaging function,

$$K(\omega) = \int_{-\infty}^{\infty} dt \varpi(t) e^{i\omega t} \quad (2.2.12)$$

which also possess the properties as $\varpi(t)$, with $K(0) = 1$ (by Eq.) and has a finite width of $\omega_K \approx 1/\Delta T$. The Fourier transform of the convolution Eq.2.2.11 is just the product of the individual Fourier transforms:

$$\overline{f(\omega)} = K(\omega) f(\omega) \quad (2.2.13)$$

The time scale is separated by imposing the condition:

$$g \ll \omega_K \ll \omega_0 \quad (2.2.14)$$

in which the first inequality is necessary to ensure the validity of time averaging, i.e. the state is slow varying at frequency g , and the latter is enforced to split the time scale between $|\omega_0 - \omega|$ with $\omega_0 + \omega$. Thus for small frequencies $\omega \ll \omega_K$ the function $f(\omega)$ will remain almost unchanged, on the contrary those with large frequencies will be strongly suppressed.

Upon applying the time-averaging to the interaction Hamiltonian 2.2.4, together with the fact that slow varying state $|\psi(t)\rangle = |\overline{\psi(t)}\rangle$ and $\overline{df/dt} = d\overline{f}/dt$, we obtain

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \overline{H}_{int} |\psi(t)\rangle \quad (2.2.15)$$

where $\overline{H}_{int}(t) = \overline{H}_{int}^{(r)}(t) + \overline{H}_{int}^{(ar)}(t)$, and since

$$\overline{e}^{\pm i(\omega_o \pm \omega_k)t} = \int_{-\infty}^{\infty} dt' \varpi(t') e^{\pm i(\omega_o \pm \omega_k)(t+t')} \quad (2.2.16)$$

$$= \int_{-\infty}^{\infty} dt' \varpi(t') e^{\pm i(\omega_o \pm \omega_k)(t')} e^{\pm i(\omega_o \pm \omega_k)t} = K(\pm(\omega_o \pm \omega_k)) e^{\pm i(\omega_o \pm \omega_k)t} \quad (2.2.17)$$

The results are

$$\overline{H}_{int}^{(r)}(t) = \hbar \sum_{\mathbf{k}} g_{\mathbf{k}}^{(r)} \left(a_{\mathbf{k}} \sigma_+ e^{i(\omega_o - \omega_k)t} + a_{\mathbf{k}}^\dagger \sigma_- e^{-i(\omega_o - \omega_k)t} \right) \quad (2.2.18)$$

and

$$\overline{H}_{int}^{(ar)}(t) = \hbar \sum_{\mathbf{k}} g_{\mathbf{k}}^{(ar)} \left(a_{\mathbf{k}} \sigma_- e^{-i(\omega_o + \omega_k)t} + a_{\mathbf{k}}^\dagger \sigma_+ e^{i(\omega_o + \omega_k)t} \right) \quad (2.2.19)$$

where we have denoted $g_{\mathbf{k}}^{(r)} = K(\omega_o - \omega_k)g_{\mathbf{k}}$ and $g_{\mathbf{k}}^{(ar)} = K(\omega_o + \omega_k)g_{\mathbf{k}}$ and used the property $K(-\omega) = K(\omega)$.

Since ω_K is conditioned by Eq.2.2.14, in the case of resonance, where ω_0 is close to ω , we have

$$|\omega_0 - \omega| \ll \omega_K \ll \omega_0 + \omega \quad (2.2.20)$$

Using the property of $K(\omega)$, it can be concluded that $\overline{H}_{int}^{(ar)}(t) \approx 0$ since $K(\omega_0 + \omega) \approx 0$.

Hence apart from the replacement of $g_{\mathbf{k}} = K(\omega)g_{\mathbf{k}}$, the full Hamiltonian after RWA is given by

$$H = \frac{1}{2} \hbar \omega_0 \sigma_z + \sum_{\mathbf{k}} \hbar \omega_k a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \hbar \sum_{\mathbf{k}} g_{\mathbf{k}}^{(r)} \left(a_{\mathbf{k}} \sigma_+ + a_{\mathbf{k}}^\dagger \sigma_- \right) \quad (2.2.21)$$

2.3 Cavity Quantum Electrodynamics

The atoms we considered previously are in free space and are allowed to interact with multimode fields. We now wish to explore a more fundamental light-matter interaction by subjecting the atom to single mode of field, which can be done by placing the atom inside a resonant cavity. This will lead us to the subject of cavity quantum electrodynamics.

namics (Cavity QED), which studies the interaction between atoms and the quantized electromagnetic modes inside a cavity.

2.3.1 Atom Cavity coupling

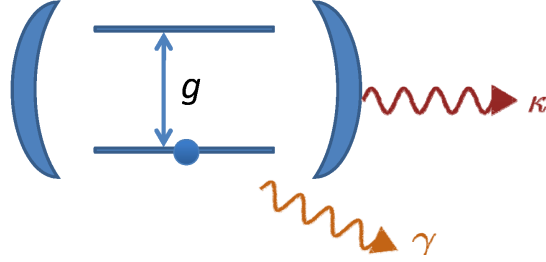


Figure 2.1: Two-level atom in a resonant cavity, described by three parameters g , κ and γ

We consider a two-level atom placed inside a cavity, as shown in Fig. 2.1. Inside the cavity, the radiation modes are discrete. We are particularly interested in the case where the transition frequency of the atom are on resonance with one of the mode of the cavity, leading to enhancement of the interaction between the atom and the field.

The key parameters that determines the relative strength of atom cavity coupling are [9]:

- Atom-photon coupling strength g
- Non resonant decay rate γ
- Cavity decay rate κ

The first parameter is the familiar coupling strength g in Hamiltonian Eq. (2.2.21) defined in previous sections. The photon decay rate is governed by the properties of the cavity, which is often characterized by the quality factor Q . κ is related to Q by

$$\kappa = \frac{\omega}{Q} \quad (2.3.1)$$

This implies that to minimize the leakage of photon from the mirrors of the cavity, a high Q factor cavity is required. The decay of the atom is due to several factors. For instance,

the atom could emit photon into modes other than cavity modes, or decay without emission.

When the parameter $g \ll \{\kappa, \gamma\}$, the system is in *weak coupling* regime and undergo irreversible process since the photons are lost from the system due to decay of both cavity and atom. Conversely, when $g \gg \{\kappa, \gamma\}$, we have *strong coupling* in which the photon is reabsorbed before escaping from the cavity, leading to reversible process. This regime of reversible light-atom interactions of single quantized mode can be described by the Jaynes-Cummings model, which will be discussed next.

2.3.2 Jaynes-Cummings model

The interaction of a single two-level atom with a single mode of the electromagnetic field is the most fundamental light-matter interaction. In this case, the field consists of single mode and our Hamiltonian Eq.2.2.21 simplifies to

$$H = H_0 + H_{\text{int}} = \frac{1}{2}\hbar\omega_0\sigma_z + \hbar\omega a^\dagger a + \hbar g^{(r)} (a\sigma_+ + a^\dagger\sigma_-) \quad (2.3.2)$$

with

$$H_0 = \frac{1}{2}\hbar\omega_0\sigma_z + \hbar\omega a^\dagger a \quad (2.3.3)$$

$$H_{\text{int}} = \hbar g^{(r)} (a\sigma_+ + a^\dagger\sigma_-) \quad (2.3.4)$$

Since $[H_0, H_{\text{int}}] = 0$, the eigenstate of H can be expressed as the linear combination of the degenerate eigenstates of the H_0 , namely $|n, e\rangle$ and $|n+1, g\rangle$ ³. They define a two-dimensional subspaces can be diagonalized separately. We have

$$H|n, e\rangle = \hbar\omega n + \frac{\hbar}{2}\omega_0|n, g\rangle + \hbar g\sqrt{n+1}|n+1, g\rangle, \quad (2.3.5)$$

$$H|n+1, g\rangle = \hbar\omega(n+1) - \frac{\hbar}{2}\omega_0|n+1, g\rangle + \hbar g\sqrt{n+1}|n, e\rangle \quad (2.3.6)$$

³ $|n, e\rangle$ is the product states of the unperturbed field and atom, $H_0, |n\rangle \otimes |e\rangle$

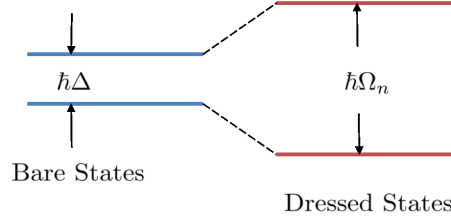


Figure 2.2: Energy level splitting due to interaction between atom and quantized field

Thus in the 2×2 subspace spanned by $|n, e\rangle$ and $|n+1, g\rangle$, the corresponding matrix representation is⁴

$$H = (n + 1/2) \hbar \omega \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{\hbar}{2} \begin{pmatrix} \Delta & 2g\sqrt{n+1} \\ 2g\sqrt{n+1} & -\Delta \end{pmatrix} \quad (2.3.7)$$

The eigenvalues and eigenstates turns out to be (See Appendix B.1),

$$E_{\pm} = (n + 1/2) \hbar \omega \pm \frac{\hbar}{2} \Omega_n \quad (2.3.8)$$

and

$$|n, +\rangle = \cos \frac{\theta}{2} |n, e\rangle + \sin \frac{\theta}{2} |n+1, g\rangle, \quad (2.3.9)$$

$$|n, -\rangle = -\sin \frac{\theta}{2} |n, e\rangle + \cos \frac{\theta}{2} |n+1, g\rangle \quad (2.3.10)$$

with

$$\Omega_n = \sqrt{\Delta^2 + 4g^2(n+1)}, \quad (2.3.11)$$

$$\theta = \tan^{-1} \left(\frac{2g\sqrt{n+1}}{\Delta} \right) \quad (2.3.12)$$

The states $|n, \pm\rangle$ are often denoted as “dressed states” or Jaynes-Cummings doublets, as shown in Fig. 2.2. When the system is on resonance, $\Delta = 0$, the above results read

$$|n, \pm\rangle = \frac{1}{\sqrt{2}} (|g, n+1\rangle \pm |n, e\rangle) \quad (2.3.13)$$

⁴For simplicity we omit the superscript on g

with eigenvalues

$$E_{\pm} = (n + 1/2) \hbar\omega \pm \hbar g\sqrt{n+1} \quad (2.3.14)$$

Hence the splitting between the eigenstates is $E_+ - E_- = 2\hbar g\sqrt{n+1}$, giving rise to anharmonicity in energy spacing as n goes up, which is often called Jaynes-Cummings ladder[9].

If the atom is initially in the excited state and the field has n photons, the probability for the atom to be in the excited state with n photons in the field at time t is then given by

$$P_e = |\langle n, e | e^{-iHt/\hbar} | n, e \rangle|^2 = \cos^2(g\sqrt{n+1}t) \quad (2.3.15)$$

which implies that if the initial state is $|0, e\rangle$, it is undergoing coherent light-atom oscillation called Rabi oscillation at frequency of g/π .

2.4 Ultrastrong coupling

In the strong coupling regime, the Jaynes-Cummings model has been shown to describe the dynamics of cavity QED in experiments such as trapped ion and Rydberg atom QED. Nowadays, solid-state semiconductor or superconductor has allowed the advent of the ultrastrong coupling regime, in which the coupling strength g is comparable or larger than appreciable fraction of the mode frequency ω , which is often quantified as $g/\omega \geq 0.1$. In this microwave regime, the RWA will no longer be valid, and the Hamiltonian becomes analytically unsolvable.

The first experimental evidence was reported in March [17], in which coupling strength of g/ω up to 12% is achieved in superconducting circuit by inductive coupling of flux qubit⁵. The group managed to observe the anticrossings spectra which results from the counterrotating terms, indicating the breakdown of RWA. Another group working on superconducting circuit also reported the observation of energy shift in energy level due to the presence of counter-rotating term called Bloch-Siegert Shift[8].

⁵More on the experimental details will be presented in Chapter 4

2.4.1 Breakdown of RWA

We wish to gain more insight on how the coupling strength, resonator frequency and the validity of RWA are related to each other. Hence we revisit the time averaging method discussed in Sec. 2.2.2. Here we'll consider the case where the system is on resonant and off resonant.

Resonant Case: $\Delta \approx 0$

When the system is near resonant, $|\omega_0 - \omega| \ll \omega_0 + \omega$, which also implies $\omega_0/\omega \approx 1$. We invoke the inequalities Eq. (2.2.14) and apply them in both strong and ultrastrong coupling regime.

1. Strong coupling

In strong coupling regime, where $g \gg \{\kappa, \gamma\}$, we enforce the time scale splitting by requiring

$$\frac{g}{\omega} \ll \frac{\omega_K}{\omega} \ll \frac{\omega_0}{\omega} \quad (2.4.1)$$

In traditional quantum optic system such as 3D optical and 3D microwave [3], the ratio g/ω is of order of 10^{-7} . Hence together with the fact that $\omega_0/\omega \approx 1$, the inequality is satisfied and Eq. (2.2.20) follows

2. Ultrastrong coupling

In this regime, the ratio g/ω approaching 0.1, by Eq.2.4.1

$$0.1 \ll \frac{\omega_K}{\omega} \ll 1 \quad (2.4.2)$$

If strict condition is imposed, the above inequality cannot be satisfied simultaneously. The first inequality must hold in order to perform time averaging, therefore the second inequality does not apply. As a result, the width (ω_K) of the time averaging function $K(\omega)$ is of the same order of magnitude as ω and ω_0 (since $\omega \approx \omega_0$). This implies that $K(\omega_0 + \omega)$ in $H_{int}^{(ar)}$ is nonzero and cannot be discarded, leading to the manifestation of the counter rotating terms in the Hamiltonian and RWA

breaks down.

Large detuning case, $\Delta \leq \omega_0 + \omega$

In the regime where $|\omega_0 - \omega|$ comparable with $\omega_0 + \omega$, which happens when either ω or ω_0 are much larger than the other, the time scale among the $|\omega_0 - \omega|$ and $\omega_0 + \omega$ cannot be split. This means $K(\omega_0 + \omega)$ is of the same order of magnitude of $K(\omega_0 - \omega)$ if time averaging were to be performed. Hence RWA cannot be applied in both case of strong and ultrastrong coupling in non-resonant case.

2.4.2 Effect arises from ultrastrong coupling

Since the normalized coupling strength g/ω in traditional optics is of the order of $10^{-6} - 10^{-7}$ only, ultrastrong coupling is not of major concern. Nevertheless, some studies had been done[20]. For instance, Bloch and Siegert(1940) showed that the counter rotating parts can give rise to a frequency shift (Bloch Siegert Shift). The shifts are small but may affect the long-time behavior of the overall system. It also claimed to lead to chaotic behavior when semiclassical treatment is applied (Milonni et al. 1983).

Recently, new interest in Jaynes Cummings model has started to emerge again with the advent of solid state QED. There are several recent theoretical studies, such as generalized rotating-wave approximation [11] which combines adiabatic approximation and standard RWA and strong dispersive regime beyond RWA by Zueco et al.[25]⁶

We explored some of the consequences arises from the inclusion of the counter-rotating term and several effects. For example, Bloch-Siegert Shift are rederived by using time independent perturbation theory. Due to time and space constraint, here we shall only present some key concepts and general ideas.

⁶This is of our interest in this project and will be discussed in great detail on next chapter.

Bloch-Siegert Shift

We perform calculation of the Bloch-Siegert shift using time independent second order perturbation theory. Calculation in terms of the uncoupled basis yields an approximate results. Using dressed state basis $|n, \pm\rangle$ from Eq. (2.3.13), we eventually obtain [21]

$$\Omega_{BS} = \frac{g^2}{4\omega} \quad (2.4.3)$$

This shift will give rise to further repulsion of the energy level in the dressed basis.

Quantum Anti-Zeno Effect

The phenomenon in which frequent measurements of a quantum-mechanical system may cause its evolution to change, slow down or freeze is known as quantum Zeno effect. Lizuain et. al. [14] shows that when periodic measurement of time δt is the fastest timescale in the system, RWA breakdown and enhancement of the Zeno or even anti-Zeno effect, in which decay of the excited state is accelerated compared with the RWA case can be observed.

Chapter 3

Qubit-Resonator dynamic in dispersive regime

One of the practical interest in qubit oscillator coupling is the dispersive limit, in which the qubit and the resonator are far detuned from each other compared to the coupling strength. This implies that the gates rely on the virtual excitation of the resonator and hence the effect from the cavity decay can be minimized. Dispersive regime in RWA has been discussed in several papers [3, 2] and NRWA regime also start to gain interest [25]. Motivated by the importance of dispersive regime and in view of the experimental tendency toward stronger qubit-oscillator coupling, we follow up the work done by Zueco et al. in [25] and derive an general expression in term of the time averaging function discussed in Section 2.2.2, which lead to both RWA and Non-RWA (NRWA) under valid assumption. We also obtained the limit of coupling strength which is dependent on initial parameters of the system¹. We further extend the discussion by investigating the properties dispersive shift, interqubits coupling strength and gate operations under the constraint of the dispersive limit.

¹Although it is mathematically straight forward, to my best knowledge, I have not found this discussion elsewhere

3.1 Dispersive regime

In dispersive regime, the qubit and the resonator are highly detuned with each other with respect to the coupling strength, $g \ll |\omega_0 - \omega|$. The main advantages in working in dispersive regime is that the interaction between the qubits and the resonator is mediated by exchange of virtual photon in the cavity. This implies that the effect from the decay of the cavity κ is reduced and the overall lifetime of the system is enhanced [3]. Hence the speed of the gate constructed will not be limited by the quality factor of the resonator. On the other hand, a non-demolition-type measurement, in which the state of the qubit can be obtained by probing the frequency of the resonator solely, is also possible in this regime.

In order for the dispersive limit and RWA to be both valid, we required

$$g \ll |\omega_0 - \omega| \ll \omega_0 + \omega \quad (3.1.1)$$

in which the first inequality comes from the dispersive limit, while the latter is required in order to deploy RWA and obtain 2.2.21. These inequalities hold when the coupling strength is not too strong.² As the recent experiments[17, 8] indicates the possibility of reaching even stronger coupling, it is possible that the second inequality is violated and RWA no longer applicable. Hence in this regime the condition 3.1.1 becomes

$$g \ll |\omega_0 - \omega| \leq \omega_0 + \omega \quad (3.1.2)$$

in which the second inequality can be shown by the triangle inequality³

3.1.1 Time Averaging in Dispersive Regime

The time averaging method can again be incorporated in the dispersive regime to determine the validity of RWA. So far we know that three inequalities must be satisfied:

²There is a distinction between strong coupling strength and strong coupling regime, in which in the first context we refer to the strength of coupling compared with the resonator strength, while the latter means $g \ll \kappa, \gamma$.

³In this context, we assumed that ω_0 and ω are always positive

1. Time averaging condition: $g \ll \omega_K$
2. Dispersive limit: $g \ll |\omega_0 - \omega|$
3. Time scale splitting: $|\omega_0 - \omega| \ll \omega_K \ll \omega_0 + \omega$

Upon combining the inequalities, we obtained:

$$g \ll |\omega_0 - \omega| \ll \omega_K \ll \omega_0 + \omega \quad (3.1.3)$$

This is the dispersive form of RWA, analogous to Eq. (2.2.14). These inequalities are automatically satisfied in strong coupling regime, in which g is relatively small. When it comes to ultrastrong coupling regime, if we wish to operate in dispersive limit and perform time average, condition (1) and (2) need to hold, the increase in g will lead to an increase in both ω_K and $|\omega_0 - \omega|$ proportionally, resulting in⁴

$$g \ll |\omega_0 - \omega| \leq \omega_0 + \omega \ll \omega_K \quad (3.1.4)$$

and we obtained the case 2.4.1 where the timescale cannot be split.

3.1.2 Constraints and Conditions

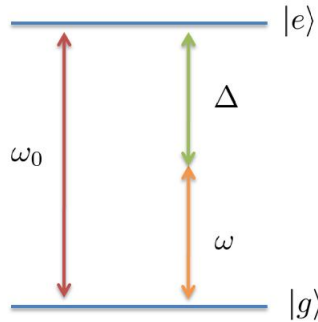


Figure 3.1: Detuning increases as we increase ω_0 or decrease ω

There are certain constraints and conditions on the inequality Eq. (3.1.2) that need to be taken into account before we move on. We noted that in order to satisfy Eq. (3.1.2),

⁴ $\omega_0 + \omega$ will either decrease or increase depending on setup

the ratio between $g/|\Delta|$, where $\Delta = \omega_0 - \omega$ must be kept at certain minimum. Hence depending on our initial conditions and variable, the magnitude of $|\omega_0 - \omega|$ and $\omega_0 + \omega$ will be changed appropriately.

Assuming the coupling strength and the qubit/resonator energy are independent of each other, we first consider $\Delta = \omega_0 - \omega > 0$ (Fig. 3.1) and set

$$\omega_0 - \omega = \alpha g \quad (3.1.5)$$

Normally α is approximately 10[2]. There are two cases to be considered.

Increase ω_0 , keeping ω constant

In this case, we have $\omega_0 + \omega = 2\omega + \alpha g$. Therefore as we further increase the coupling strength, ω_0 will increase, leading to

$$\frac{\omega_0 - \omega}{\omega_0 + \omega} = \frac{\alpha g}{2\omega + \alpha g} \approx 1 \quad (3.1.6)$$

At this domain RWA no longer holds, although there exists finite constant difference between $|\omega_0 - \omega|$ and $\omega_0 + \omega$, i.e. $\omega_0 + \omega - (\omega_0 - \omega) = 2\omega_0$, which is illustrated below (Fig. 3.2)

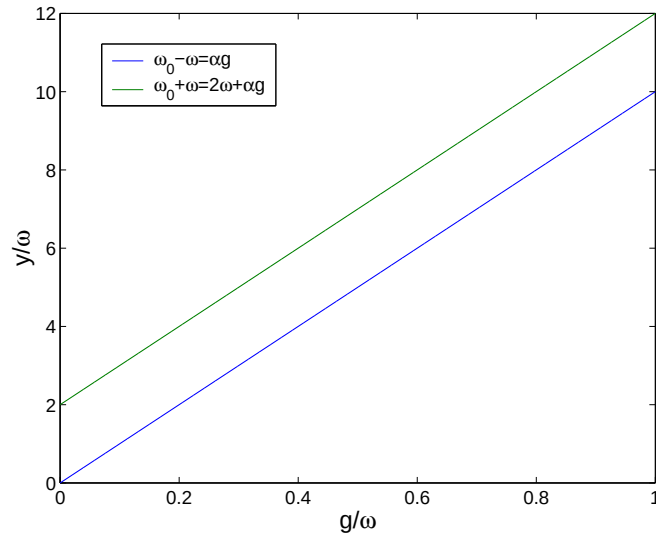


Figure 3.2: $\omega_0 + \omega$ and $\omega_0 - \omega$ increases as g increase, keeping ω constant.

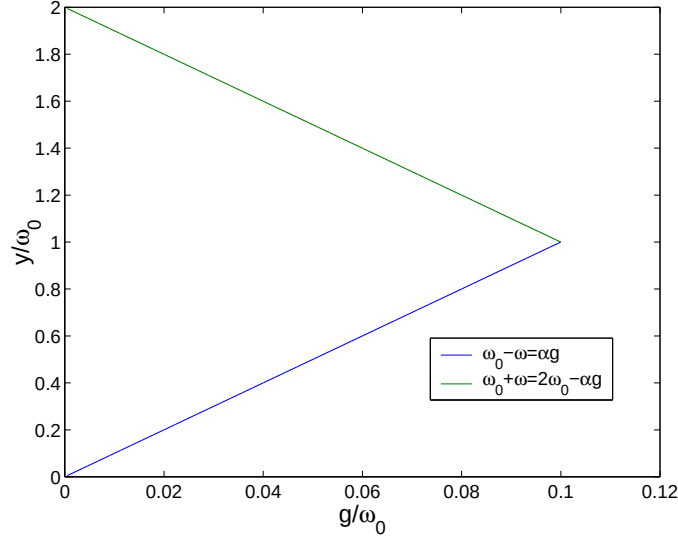


Figure 3.3: $\omega_0 + \omega$ and $\omega_0 - \omega$ approaching each other as g increase, keeping ω_0 constant.

Decrease ω , keeping ω_0 constant

This time we'll try to satisfy the inequality by decreasing ω instead as the coupling strength increase. However this will lead to a different situation as $\omega_0 + \omega$ will be decreased as well as ω is getting smaller. Eventually the $\omega_0 - \omega$ and $\omega_0 + \omega$ will coincidence, which imposed a maximum bound for the coupling strength in dispersive limit for this case

$$\omega_0 - \omega = \omega_0 = \alpha g \quad (3.1.7)$$

$$\frac{g}{\omega_0} \leq \frac{1}{\alpha} \quad (3.1.8)$$

The corresponding curve with $\alpha = 0.1$ is shown in Fig. 3.3. At $g/\omega \approx 0.1$, the resonator is effectively turned off ($\omega \approx 0$), hence further increase in coupling strength is meaningless in this sense. When $\Delta = \omega_0 - \omega < 0$, the whole analysis above can be repeated by interchanging ω with ω_0 , however it has to be taken note that in this situation there exists a bound for the increase in ω due to the existence of subsequent higher energy level.

With this, we are now ready to work out a rough estimation of the breakdown point of RWA. Assuming we are tuning ω_0 and keeping ω constant, by dividing Eq. (3.1.3) with ω gives

$$\frac{g}{\omega} \ll \frac{\omega_0}{\omega} - 1 \ll \frac{\omega_K}{\omega} \ll \frac{\omega_0}{\omega} + 1 \quad (3.1.9)$$

For instance, if we set $\frac{\omega_0+\omega}{\omega_0-\omega} = \beta$,

$$\frac{\omega_0}{\omega} + 1 = \beta \left(\frac{\omega_0}{\omega} - 1 \right) \quad (3.1.10)$$

$$\frac{\omega_0}{\omega} = \frac{1 + \beta}{\beta - 1} \quad (3.1.11)$$

By Eq. (3.1.5), $\frac{\omega_0}{\omega} - 1 = \alpha \frac{g}{\omega}$. Substituting into Eq. 3.1.11, we eventually obtain

$$\frac{g}{\omega} = \frac{2}{\alpha(\beta - 1)} \quad (3.1.12)$$

Putting $\alpha = 10, \beta = 10$ in the equation, we obtain ratio $g/\omega \approx 0.02$, which is in the strong coupling RWA regime. However when $g/\omega \approx 0.1$, known as ultrastrong coupling regime, $\beta \approx 3$, indicating close ratio between $\omega_0 - \omega$ and $\omega_0 + \omega$ where RWA fails. Similar analysis can be done on case C.3.2, giving

$$\frac{g}{\omega_0} = \frac{2}{\gamma(\beta + 1)} \quad (3.1.13)$$

in which when $\beta = 1$, i.e. $\omega = 0$ will resulting in 0.1 for g/ω_0 , which is the intersection point in Fig. 3.3.

3.2 Dispersive Regime in One Qubit

3.2.1 General Expression

We will now work out the dispersive regime in general, by incorporating the time averaging functions introduced in Sec. 2.2.2 to keep track of both the co-rotating and counter-rotating term in the Hamiltonian. The full atom-field interaction, often denoted as *Rabi Model* is

$$H = H_0 + H_I = \frac{\hbar}{2}\omega_0\sigma_z + \hbar\omega a^\dagger a + \hbar g (a + a^\dagger) \sigma_x \quad (3.2.1)$$

where $H_0 = \frac{\hbar}{2}\omega_0\sigma_z + \hbar\omega a^\dagger a$, $H_I = \hbar g(a + a^\dagger)\sigma_x$ and $\sigma_x = \sigma_+ + \sigma_-$ ⁵. Once again we split the interaction Hamiltonian to co-rotating and counter-rotating parts

$$H_I = \hbar g^{(r)}X_+ + \hbar g^{(ar)}Y_+ \quad (3.2.2)$$

with

$$X_\pm = a\sigma_+ \pm a^\dagger\sigma_-, \quad (3.2.3)$$

$$Y_\pm = a\sigma_- \pm a^\dagger\sigma_+ \quad (3.2.4)$$

and it is reminded that $g^{(r)} = gK(\omega_0 - \omega)$, $g^{(ar)} = gK(\omega_0 + \omega)$, where $K(\omega)$ is the time-averaging function. Eventually the Hamiltonian takes the form

$$H = \frac{\hbar}{2}\omega_0\sigma_z + \hbar\omega a^\dagger a + \hbar g^{(r)}X_+ + \hbar g^{(ar)}Y_+ \quad (3.2.5)$$

In the dispersive limit determined by Eq. (3.1.2), the Hamiltonian can be diagonalized approximately. By introducing the unitary transformation,

$$U = \exp(\lambda_\Delta X_- + \lambda_\Sigma Y_-) \quad (3.2.6)$$

where $\lambda_\Delta = g^{(r)}/\Delta$, $\lambda_\Sigma = g^{(ar)}/\Sigma$ with $\Delta = \omega_0 - \omega$ and $\Sigma = \omega_0 + \omega$, the Hamiltonian can be transformed up to second order in λ_Δ and λ_Σ :

$$\begin{aligned} H_{\text{disp}} &= U^\dagger H U \quad (3.2.7) \\ &= H + [H, \lambda_\Delta X_- + \lambda_\Sigma Y_-] + \frac{1}{2}[[H, U], U] \\ &= H + \lambda_\Delta [H, X_-] + \lambda_\Sigma [H, Y_-] \\ &\quad + \frac{1}{2} \left(\lambda_\Sigma^2 [[H, X_-], X_-] + \lambda_\Delta \lambda_\Sigma [[H, X_-], Y_-] + \lambda_\Delta^2 [[H, Y_-], Y_-] + \lambda_\Sigma \lambda_\Delta [[H, Y_-], X_-] \right) \end{aligned}$$

⁵A more general case, with σ_x replaced by $\cos\theta\sigma_z - \sin\theta\sigma_x$. Nevertheless the σ_z term can be discarded by applying polaron transformation [18]

Despite being one of the major work in this project, for consistency the detail of the derivation will be given in the appendix C.1 instead. After tedious calculation, we obtain:

$$H_{\text{disp}} = H_0 + \frac{\hbar}{2} \left(\frac{g^{(r)2}}{\Delta} + \frac{g^{(ar)2}}{\Sigma} \right) (\sigma_z (2a^\dagger a + 1)) \quad (3.2.8)$$

$$+ \frac{\hbar}{2} \left(\frac{g^{(r)}g^{(ar)}}{\Delta} + \frac{g^{(ar)}g^{(r)}}{\Sigma} \right) (\sigma_z ((a^\dagger)^2 + a^2))$$

with constant terms dropped.

3.2.2 Case studies: RWA and NRWA

RWA in one qubit

In the strong coupling regime where RWA applies (eg. $g/\omega \approx 0.01$), counter-rotating term can be discarded in first order, which give rise to

$$H_{\text{RWA,disp}} = H_0 + \frac{\hbar}{2} \left(\frac{g^{(r)2}}{\Delta} \right) [\sigma_z (2a^\dagger a + 1)] \quad (3.2.9)$$

Upon arranging and set $g^{(r)} = g$, we obtain

$$H_{\text{RWA,disp}} = \frac{\hbar}{2} \left(\omega_0 + \frac{g^2}{\Delta} \right) \sigma_z + \hbar \left(\omega + \frac{g^2}{\Delta} \sigma_z \right) a^\dagger a \quad (3.2.10)$$

We may interpret this expression as the AC Stark shift[3, 25], in which in our case is a qubit dependent shift due to the presence of σ_z in the expression proportional to $a^\dagger a$. Hence we have

$$\omega \rightarrow \omega_{\text{RWA}} = \omega \pm g^2/\Delta \quad (3.2.11)$$

where the sign depends on the state of the qubit as shown in Fig. 3.4. On the other hand, the fact that the resulting Hamiltonian $H_{\text{RWA,disp}}$ commutes with σ_z , i.e. $[H_{\text{RWA,disp}}, \sigma_z] = 0$ allows quantum nondemolition measurement since the qubit states will not be changed upon the evolution of the system. Hence the state of the qubit can be inferred by probing the resonator frequencies.

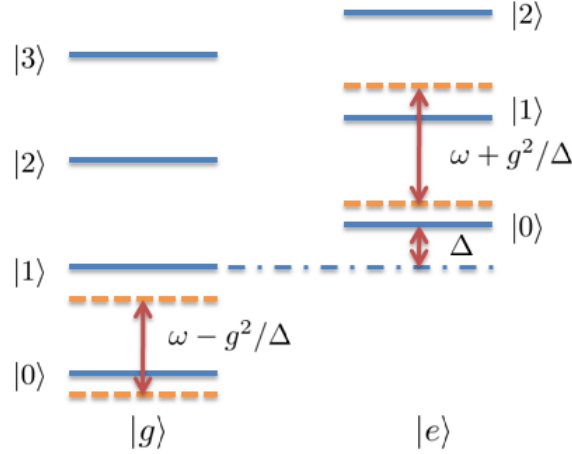


Figure 3.4: Energy spectrum in the dispersive regime (Dashed Line). The level separation is independent of n but is qubit state dependent.

NRWA in one qubit

For ultrastrong coupling, all the term in Eq. (3.2.8) will be retained, giving the expression

$$H_{\text{NRWA,disp}} = \frac{\hbar}{2} \left[\omega_0 + g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \right] \sigma_z + \hbar \left[\omega + \frac{g^2}{2} \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \sigma_z \right] (a + a^\dagger)^2 \quad (3.2.12)$$

This expression is analogous to the RWA expression 3.2.10, with an extra contribution of λ_Σ in the coupling term⁶. However, this Hamiltonian is not diagonal in the eigenbasis of H_0 due to $(a^\dagger)^2$ and a^2 . Nevertheless, for not too small ω , we can truncate these terms with RWA argument since they are of the second order [18]. In the interaction picture, they transformed as $a^2(t) = a^2 e^{2i\omega t}$ and $(a^\dagger)^2(t) = (a^\dagger)^2 e^{-2i\omega t}$, which will be canceled by the method of averaging discussed before. We thus have

$$H_{\text{NRWA,disp}} \approx \frac{\hbar}{2} \left[\omega_0 + g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \right] \sigma_z + \hbar \left[\omega + g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \sigma_z \right] a^\dagger a \quad (3.2.13)$$

Hence analogous to RWA Hamiltonian, NRWA also give rise to shift in the oscillator frequency

$$\omega \rightarrow \omega_{\text{NRWA}} = \omega \pm g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \quad (3.2.14)$$

⁶Comparing with the expression in [25], we believe that there is typo in the transformation and the final expression, which missed out dispersive shift in the qubit term.

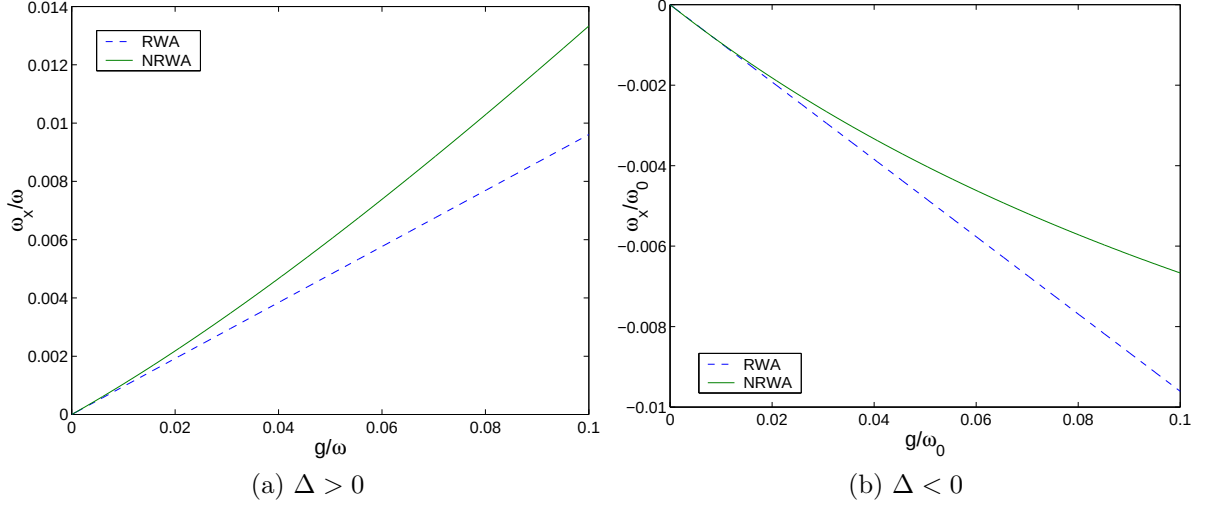


Figure 3.5: Resonator frequency shift as coupling strength increases, with (a) ω_0 increase, ω constant (b) ω increased, ω_0 constant

which implies that dispersive readout is also possible even in ultrastrong coupling regime. This result agreed with the result in [25] with different argument.

We can examine the frequency shift more rigorously by applying the constraints mentioned in Sec. 3.1.2. When we increase ω_0 or ω in positive and negative detuning respectively, RWA and NRWA predicted shift as Fig. 3.5.⁷ It can be seen clearly RWA result overestimates the frequency shift for negative detuning $\Delta < 0$, and underestimates when it comes to positive detuning. It is more interesting when we decrease ω_0 or ω in positive and negative detuning. In the case of decreasing ω , it can be seen that the shift increased quadratically. This is reasonable in this regime as the effect of the energy level of atom, or $\hbar\omega_0$ on the frequency shift is more apparent due to larger qubit-resonator frequency ratio as ω getting smaller. On the other hand, in the limit of $\Delta \rightarrow -\omega$, which corresponds to a vanishing qubit splitting $\omega_0 \rightarrow 0$ (Fig. 3.6b), it is expected to have no shift in the resonator frequency as the qubit are “invisible”. The NRWA curves correctly predicted this, but the RWA result leads to a unreasonable frequency shift [25], indicating the breakdown of RWA.

⁷The details of the expression plotted are given in appendix C.3

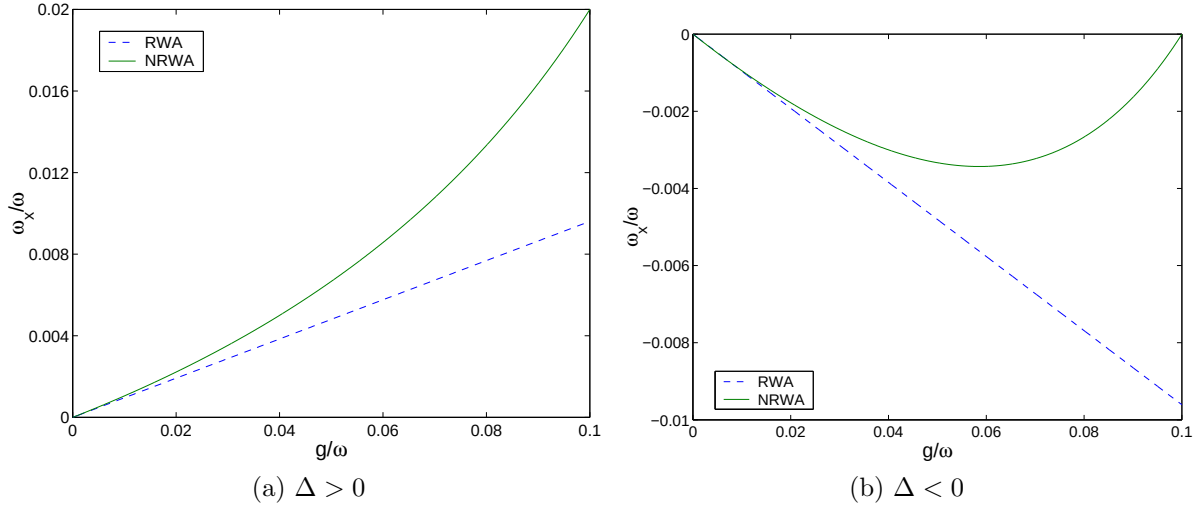


Figure 3.6: Resonator frequency shift as coupling strength increases, with
(a) ω decreased, ω_0 constant (b) ω_0 decreased, ω constant

3.3 Dispersive Regime in Two Qubit

3.3.1 General Expression

Having familiarize ourselves with the 1 qubit case, we shall now extend our discussion to multiple qubit coupled to a single mode resonator. Following the 1 qubit case, the Hamiltonian takes the form [2]

$$H = \frac{\hbar}{2} \sum_j \omega_0^j \sigma_z^j + \hbar \omega a^\dagger a + \hbar g_j (a + a^\dagger) \sigma_x \quad (3.3.1)$$

Once again we insert the time averaging function into the equation and obtain

$$H = \frac{\hbar}{2} \sum_j \omega_0^j \sigma_z^j + \hbar \omega a^\dagger a + \hbar g_j^{(r)} X_+^j + \hbar g_j^{(ar)} Y_+^j \quad (3.3.2)$$

where $X_\pm^j = a\sigma_+^j \pm a^\dagger\sigma_-^j$ and $Y_\pm^j = a\sigma_-^j \pm a^\dagger\sigma_+^j$ [cf. Eq.3.2.3]. The dispersive Hamiltonian can be obtained by the unitary transformation

$$U = \exp \left(\sum_j \lambda_\Delta^j X_-^j + \lambda_\Sigma^j Y_-^j \right) \quad (3.3.3)$$

which cancels the term in the first order of g . Following the procedure in the Sec. 3.2.1, we expand the terms up to second order in X_-^j and Y_-^j . Apart from qubit resonator coupling, there will be qubit-qubit coupling mediated by the resonator as well⁸ The resulting dispersive Hamiltonian, which constitute the one of the main result of this project, reads as

$$\begin{aligned}
 H_{\text{N,disp}} = U^\dagger H U = & \sum_j H_{\text{disp}}^j \\
 & + \sum_{j>k} \frac{\hbar}{2} \left\{ \left[g_j^{(r)} g_k^{(r)} \left(\frac{1}{\Delta_j} + \frac{1}{\Delta_k} \right) - g_j^{(ar)} g_k^{(ar)} \left(\frac{1}{\Sigma_j} + \frac{1}{\Sigma_k} \right) \right] (\sigma_-^j \sigma_+^k + \sigma_+^j \sigma_-^k) \right. \\
 & \left. + \left[g_j^{(r)} g_k^{(ar)} \left(\frac{1}{\Delta_j} - \frac{1}{\Sigma_k} \right) + g_j^{(ar)} g_k^{(r)} \left(\frac{1}{\Delta_k} - \frac{1}{\Sigma_j} \right) \right] (\sigma_-^j \sigma_-^k + \sigma_+^j \sigma_+^k) \right\}
 \end{aligned} \tag{3.3.4}$$

where H_{disp}^j is Eq. (3.2.8) for j th qubit. The last term(s) is the effective coupling strength between the qubits mediated by the resonator.

3.3.2 Case studies:RWA and NRWA

In the situation where the qubits are strongly detuned from each other, the effective coupling strength is essentially turned off since it does not conserve energy[15, 2]. This can be seen by going into the interaction picture in which each qubit are in the frame rotating at ω_0^j . In this frame, when the qubit are strongly detuned, the interaction term, for instance $\sigma_+^j \sigma_-^k e^{i(\omega_0^j - \omega_0^k)t}$ is oscillating rapidly and averages out (See Sec. 2.2.2), so the effective coupling is for all practical purposes turned off. Therefore, for maximum effective coupling, here we discuss the case where the qubits are degenerate with each other, $\omega_{0j} = \omega_{0k}$ and hence $\Delta_j = \Delta_k = \Delta$ and $\Sigma_j = \Sigma_k = \Delta$

⁸As a result, the commutator exercises is rather lengthy and involved, hence we shall exempt the reader from this suffering and include them in the appendix.

RWA in 2 qubits

In the RWA case on a 2 qubits system, the counter-rotating term is discarded in the first place. By setting $g_j^{(r)} = g_k^{(r)} = g$, we obtain

$$H_{\text{RWA,disp}} = \frac{\hbar}{2} \sum_j \left(\omega_{0j} + \frac{g^2}{\Delta} \right) \sigma_z^j + \hbar \sum_j \left(\omega + \frac{g^2}{\Delta} \sigma_z^j \right) a^\dagger a + \hbar \frac{g^2}{\Delta} (\sigma_-^j \sigma_+^k + \sigma_+^j \sigma_-^k) \quad (3.3.5)$$

which agrees with expression in [3]. The interqubit coupling strength of RWA again takes the form g^2/Δ , which we shall denote as

$$J_{\text{RWA}} = \frac{g^2}{\Delta} \quad (3.3.6)$$

Since the ratio g/Δ need to be kept constant to satisfy the inequality 3.1.2, we may rewrite $J_{\text{RWA}} = g \times (g/\Delta)$. Hence the interqubit coupling strength will increase linearly in the regime of RWA. The last term in the above Hamiltonian is the so called isotropic XY interaction [4], which can be used to construct two qubit gates. In matrix representation, it is represented by the unitary operator,

$$U_{\text{RWA}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos J_{\text{RWA}} t & i \sin J_{\text{RWA}} t & 0 \\ 0 & i \sin J_{\text{RWA}} t & \cos J_{\text{RWA}} t & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.3.7)$$

For instance, by turning on the coupling for $t = \pi\Delta/4g^2$, we can generate a $\sqrt{i\text{SWAP}}$ gate which can be used to transform the state $|e_j, g_k\rangle$ into an entangled state $1/\sqrt{2}(|e_j, g_k\rangle + i|g_j, e_k\rangle)$.⁹

⁹The details on the gate operation is supplied in appendix B.2

NRWA in 2 qubits

In ultrastrong coupling regime, all the terms will be retained and by setting all g to be equal we obtain

$$\begin{aligned}
H_{\text{NRWA,disp}} = & \frac{\hbar}{2} \sum_j \left[\omega_0 + g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \right] \sigma_z^j \\
& + \hbar \sum_j \left[\omega + \frac{g^2}{2} \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) \sigma_z^j \right] (a + a^\dagger)^2 \\
& + \hbar g^2 \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) (\sigma_x^j \sigma_x^k)
\end{aligned} \tag{3.3.8}$$

with $\sigma_x^j \sigma_x^k = (\sigma_-^j \sigma_+^k + \sigma_+^j \sigma_-^k + H.c.)^{10}$. We observed that this result is indeed the generalization of Eq. (3.2.12), with an additional qubit-qubit coupling term. However, rather than just renormalization of parameters, the NRWA result give rise to an Ising type $\sigma_x^j \sigma_x^k$ coupling rather than isotropic X-Y interaction predicted by RWA, in which its unitary operator is given by,

$$U_{\text{NRWA}} = \begin{pmatrix} \cos J_{\text{NRWA}} t & 0 & 0 & i \sin J_{\text{NRWA}} t \\ 0 & \cos J_{\text{NRWA}} t & i \sin J_{\text{NRWA}} t & 0 \\ 0 & i \sin J_{\text{NRWA}} t & \cos J_{\text{NRWA}} t & 0 \\ i \sin J_{\text{NRWA}} t & 0 & 0 & \cos J_{\text{NRWA}} t \end{pmatrix} \tag{3.3.9}$$

where we have

$$J_{\text{NRWA}} = g^2 \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) \tag{3.3.10}$$

This gives rise to different spectral in the construction of the qubit gates since the ground state is no longer an eigenstate (See Appendix B.2).

On the other hand, this exact result on the interqubit coupling strength at first seems to be counter intuitive, because the negative sign in front of the coefficient of $1/\Sigma$ indicates that the interqubit coupling strength decreases as the coupling strength g increases! However this paradox can be resolved by simply setting $\Delta \rightarrow \omega_0$ (turning off the resonator)

¹⁰H.c. stands for Hermitian conjugate

which results in no effective interaction between the qubits, indeed. The RWA result, by contrast still give rise to coupling between qubit which is not sensible in this case.

As a remark, for all practical purposes, ω_0 will be finite, which allows us to perform second order RWA on the NRWA result [18] to discard terms like $\sigma_+^j \sigma_+^k e^{i2\omega t}$

$$H_{\text{NRWA,disp}} \approx \frac{\hbar}{2} \sum_j \left[\omega_0 + g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \right] \sigma_z^j \quad (3.3.11)$$

$$+ \hbar \sum_j \left[\omega + \frac{g^2}{2} \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \sigma_z^j \right] (a + a^\dagger)^2 + \hbar g^2 \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) (\sigma_-^j \sigma_+^k + \sigma_+^j \sigma_-^k)$$

With this, once again we can compare results between RWA and NRWA more closely by investigating the interqubit coupling strength under the constraints mentioned in Sec. 3.1.2.

We first consider the case of positive detuning $\Delta > 0$. As the coupling strength increases, we will have either increase in ω_0 or decrease in ω . The result of interqubit coupling strength in RWA and NRWA is plotted in Fig. 3.7.¹¹

The linear behavior of RWA is exhibited, and it is shown that it gives accurate description only up to normalized coupling strength of 0.01 in which beyond this point we can see observable deviation between two curves of RWA and NRWA. In the first case we see that J_{NRWA} approaching constant even if the coupling strength g/ω_0 is being increased. This is due to the fact that difference between two term in J_{NRWA} becomes a constant as g goes further up.

The second plot illustrates the fact that the J should be zero when we reach the limit of $g/\omega_0 = 1/\alpha$ (Eq. (3.1.8)). In this situation, the interqubit coupling strength has a maximum at intermediate strength of coupling, and starts to decrease once we continue to increase the qubit-resonator coupling strength, in contrast to the linear increase in

¹¹The details of the expression plotted are given in appendix C.3

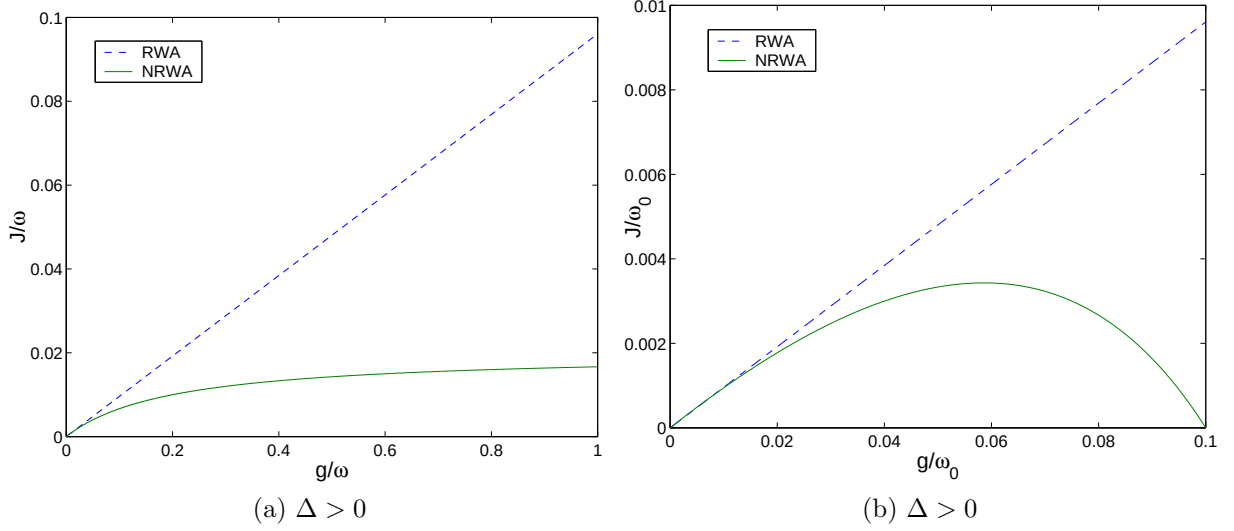


Figure 3.7: Interqubit coupling strength as coupling strength g increases, with (a) increase in ω_0 , ω constant (b) decrease in ω , ω_0 constant

J_{RWA} . For the negative detuning, we have

$$J_{\text{NRWA}} = g^2 \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) = -g^2 \left(\frac{1}{\omega - \omega_0} + \frac{1}{\Sigma} \right) \quad (3.3.12)$$

where the term arises from counter-rotating has the same sign as the RWA term $1/(\omega - \omega_0)$. Hence, increase in coupling strength will lead to an overall increase in $J_{\text{RWA/NRWA}}$ ¹². When we increase ω alone, the interqubit coupling strength increase in NRWA is linear as well, but with higher rate compared to the RWA case, which is preferable since this implies more gates operations can be done before the system undergo decoherence. However, it is not feasible to tune ω too high as this will lead to the breakdown of two level approximation due to resonance of next higher qubit energy level. Meanwhile, if we decrease ω_0 instead, the increase in interqubit strength is quadratic, but there exists a bound for the g where qubit splitting vanishes (c.f. 3.1.8).

In summary, RWA always give rise to increase in J as g increases, regardless of the initial conditions and coupling strength. This means RWA fails to capture the fact the interqubit coupling strength is subjected to the initial conditions and coupling strength as

¹²In plotting curves for negative detuning, the overall minus sign is not included since it corresponds to reverse direction of qubit rotation

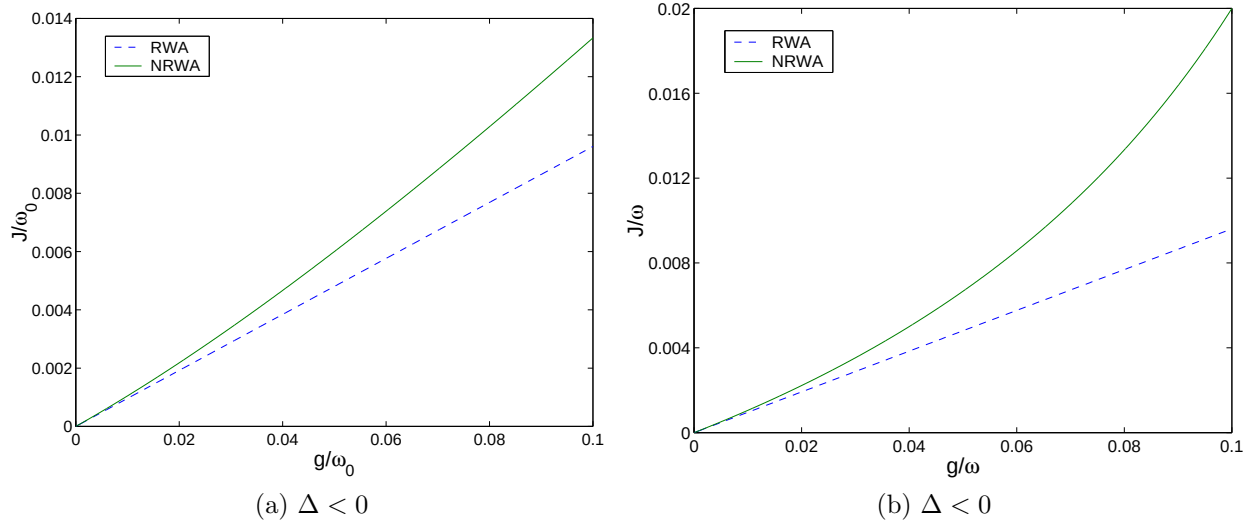


Figure 3.8: Interqubit coupling strength as coupling strength g increases, with (a) increase in ω , ω_0 constant (b) decrease in ω_0 , ω constant

predicted by the exact solution J_{NRWA} — indication of the breakdown of RWA in ultrastrong dispersive regime.

3.3.3 Intermediate RWA (IRWA)

We present the Hamiltonian in the form of two limits, which is strong coupling and ultrastrong coupling where RWA and NRWA applies respectively. One might start to ponder if there is any form of approximation that is possible to describe something in between. Here we present a form of approximation which we termed as intermediate rotating wave approximation (IRWA) which seeks to fill up this gap.

The basic ideas of the approximation is that, instead of taking to the limit of RWA/NRWA from Eq.(3.3.4), the most general expression for interqubit coupling, we insert the time averaging function into the expression $g^{(r)}$ and $g^{(ar)}$ ¹³. Hence in term of the unitary operator, the interqubit coupling Hamiltonian will exhibit transition from Eq. (3.3.7) to Eq. (3.3.9). If we express the interqubit coupling part of Eq. (3.3.4) to its unitary form, it

¹³Here we have chosen a Gaussian function which satisfied the properties mentioned in Sec. 2.2.2

reads as

$$U_{\text{IRWA}} = \begin{pmatrix} \cos J_2 t & 0 & 0 & i \sin J_2 t \\ 0 & \cos J_1 t & i \sin J_1 t & 0 \\ 0 & i \sin J_1 t & \cos J_1 t & 0 \\ i \sin J_2 t & 0 & 0 & \cos J_2 t \end{pmatrix} \quad (3.3.13)$$

where

$$J_1 = g_j^{(r)} g_k^{(r)} \left(\frac{1}{\Delta_j} + \frac{1}{\Delta_k} \right) - g_j^{(ar)} g_k^{(ar)} \left(\frac{1}{\Sigma_j} + \frac{1}{\Sigma_k} \right) \quad (3.3.14)$$

$$J_2 = g_j^{(r)} g_k^{(ar)} \left(\frac{1}{\Delta_j} - \frac{1}{\Sigma_k} \right) + g_j^{(ar)} g_k^{(r)} \left(\frac{1}{\Delta_k} - \frac{1}{\Sigma_j} \right) \quad (3.3.15)$$

With the constraints under the consideration, here we show one of the examples of the IRWA plot, which is the case where detuning Δ is positive and we decrease ω , keeping ω_0 constant.

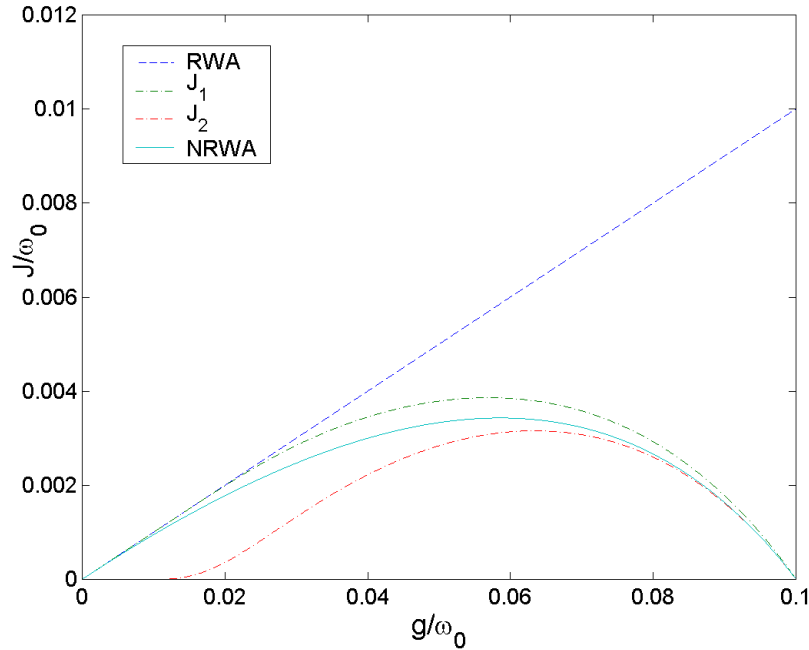


Figure 3.9: Intermediate RWA, showing the transition from RWA to NRWA regime

As we can see, when the coupling strength is small, the IRWA curve for J_1 is closer to the RWA curve and gives correction to it. As for the value J_2 , it is essentially zero in the RWA regime, and only started to manifest once the coupling get stronger, leading

to different form of the operator. As the coupling strength increases, our IRWA curves start to deviate from the RWA curve and agree better with the NRWA curve, showing the transitions from RWA to Non-RWA.

As a remark, there are several aspect which still can be explored here. For instance the relation between the speed of the measurement [14] and the width of the time averaging function ω_K , which indirectly defines the timescale involved. It might also be of interest to have different coupling strength between the qubit. For instance by having one qubit in RWA regime and another in the NRWA regime.

Chapter 4

Superconducting Circuits

After studied the dynamic of dispersive qubit-resonator system, we shall now move on to second part of the project, which is the realization of the “strong dispersive regime” in the superconducting circuits that is only accessible in a few experiments with Rydberg atoms previously[10]. In this chapter, we shall give a brief yet sufficient review on basic features on superconducting circuits. Excellent review and discussion on the subject may be found in Ref. [2, 6, 7, 24]

4.1 Josephson Effect

As mentioned in introduction, superconducting qubits are ”artificial” atom that plays the role of two level system in natural atom, yet with stronger coupling strength and tunability, which is an important advantage [24]. The superconducting circuit are macroscopic in size but exhibit generic quantum properties such as quantized energy levels and superposition of states. Their quantum states can be manipulated by parameters such as the magnetic flux or electric charge of the system. The reason behind the macroscopic quantum properties of the superconducting circuits is well explained by the theory of Bradeen, Cooper and Schrieffer (BCS theory)[1]. It states that in superconducting phase (for instance below 1.2K for aluminum), pairs of electron called *Cooper pairs* condense

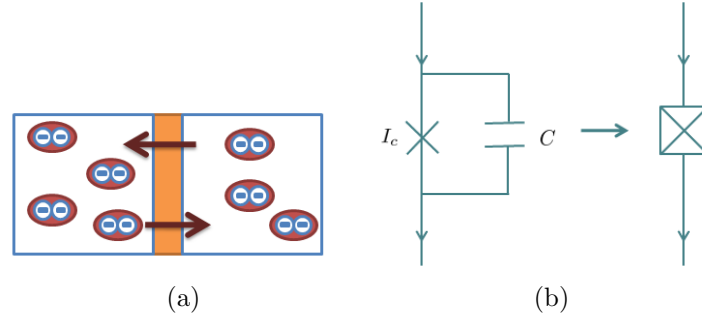


Figure 4.1: Josephson junction as a nonlinear circuit element (a) Tunnelling of the Cooper pairs (b) the schematic representation of a Josephson Junction, which also possess an intrinsic capacitance C and critical current I_c

into a single bulk macroscopic states described by the wavefunction $\Psi(\mathbf{r}, t)$ with phase ϕ

$$\Psi(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|e^{i\phi(\mathbf{r}, t)} \quad (4.1.1)$$

This macroscopic wavefunction leads to two phenomena, which are *flux quantization* and *Josephson tunneling*.

Flux Quantization

When a closed ring made up of superconducting material is penetrated by a magnetic field, the magnetic flux maintained by the circulating supercurrent in the ring will be in the flux quantum of

$$\Phi_0 = \frac{h}{2e} \quad (4.1.2)$$

which equals to $2.067 \times 10^{-15} \text{Wb}$. This quantization arises from the fact that $\Psi(\mathbf{r}, t)$ must be single valued, that is the phase change of the wavefunction after one loop must be in the integer multiple of 2π .¹ Φ_0 can be interpreted as the dual of the Cooper pair with charge $2e$.

Josephson tunnelling

The main electrical component in a superconducting circuit is the Josephson junction, which consists of two superconductors separated by an thin insulating barrier (usually

¹This phenomena is closely related to the magnetic Aharonov-Bohm effect.

2-3nm), which allows the tunneling of Cooper pairs through the barrier. Together with the property of superconducting, this give rise to the flow of current without dissipation. In 1962, Josephson discovered how the phase difference between two superconductors $\varphi = \phi_1 - \phi_2$ relates to the current and the voltage across the junction. The first one is the *current-phase relation*, in which the supercurrent through the barrier is

$$I = I_c \sin \varphi \quad (4.1.3)$$

Here I_c is the critical current of the junction, which depends on the superconducting material and construction. Next we have the AC Josephson effect, which tell us that the phase difference evolves in time as

$$V = \frac{\hbar}{2e} \frac{d\varphi}{dt} = \frac{\Phi_0}{2\pi} \frac{d\varphi}{dt} \quad (4.1.4)$$

where Φ_0 is the flux quantum defined earlier.

By using the definition of branch flux in Eq. (A.3), $\Phi(t) = \int dt' V(t')$, and the second Josephson relation, we find that

$$\Phi = \Phi_0 \frac{\varphi}{2\pi} \quad (4.1.5)$$

which in turn relates the magnetic flux Φ to the phase difference φ . With this new defination, the Josephson relations can be reexpress as

$$I = I_c \sin 2\pi \frac{\Phi}{\Phi_0} \quad (4.1.6)$$

$$|V| = d\Phi/dt \quad (4.1.7)$$

in which the latter is the familiar form of Faraday's law of induction. The inductance of the Josephson junction L_J can also be obtained from another from of Faraday's law $V = -L\dot{I}$ plus Eq. (4.1.6), giving

$$|L_J| = \frac{\Phi_0}{2\pi I_0 \cos \varphi} \quad (4.1.8)$$

This non-linear inductance introduce anharmonicity to the circuit, thus creating unequal spacing between energy level so that the lowest levels can be addressed.[24]

The quantization of the circuit is illustrated with an example of the LC circuit worked out in appendix A.3, which give rise to two relevant operator: phase ϕ and number of Cooper pairs N , satisfying the commutator relation $[\phi, N] = i$. Following the procedure in appendix A.3, with definition Eq. (4.1.3) and (4.1.4) and replace the inductance with Josephson junction, the Hamiltonian of the Josephson junction takes the form

$$H = \frac{1}{2C}P^2 + \frac{\Phi_0}{2\pi}I_c \left(1 - \cos 2\pi \frac{\Phi}{\Phi_0} \right) \quad (4.1.9)$$

By identifying

$$E_C = \frac{(2e)^2}{2C} \quad (4.1.10)$$

$$E_J = \frac{\hbar}{2e}I_c \quad (4.1.11)$$

where E_C is the charging energy of the Josephson capacitance and E_J , which can be derived by the definition $E_J = \int I(t')V(t')dt'$, is associated with the Josephson coupling energy.[6]. Together with the definition of $P = 2eN$ and $\phi = 2\pi\Phi/\Phi_0$, the Hamiltonian of the Josephson junction takes a simple form of

$$H = E_C N + E_J \cos \phi \quad (4.1.12)$$

These two energy scales are the two important parameters that determine the quantum mechanical behavior of a Josephson junction circuit.

As a remark, quantum effect are usually not observed in the regular LC circuit due to two reason: *thermal population* and *dissipation*. For the preparation of ground state, the temperature T need to satisfy:

$$k_B T \ll \hbar\omega \quad (4.1.13)$$

to prevent excitation due to thermal fluctuation. Clearly for a microwave regime device (GHz) this cannot be satisfied under room temperature. Fortunately by using a dilution refrigerator, temperatures of $T \approx 10mK$ is attainable. Hence this set the limit of the minimum operating frequencies of the system. For example $h \times 2GHz = 10 \times k_B \times 10mK$, or equivalently 1K corresponds to 20GHz[7].

4.2 Superconducting Qubits

In different regime of E_J/E_C and E_J , different flavour of superconducting qubits can be realized. Here we mainly focus on two regime, namely charge regime, $E_C \gg E_J$, which gives *charge qubit* and *flux qubit* which is in the phase regime.

4.2.1 Charge Qubit

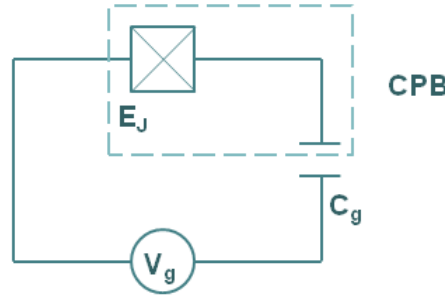


Figure 4.2: A Charge Qubit controlled by the gate voltage V_g

In this regime, the number of cooper pairs N is well defined and the phase ϕ fluctuates strongly, hence the basis state in the charge qubit is the Cooper-pair states $|N\rangle$. The charge qubit mainly consists of a small superconducting island known as a Cooper-pair box (CPB), which is driven by an applied voltage V_g through the gate capacitance C_g to induce an offset charge of $2en = C_g V_g$ (Fig. 4.2). Following steps in the appendix one can obtain

$$H = E_C(N - N_g) + E_J \cos \phi \quad (4.2.1)$$

In this setup $E_C = (2e)^2/2(C_J + C_g)$ in which the gate capacitance need to be taken into account. In the case where quasiparticle tunneling is suppressed at low temperate

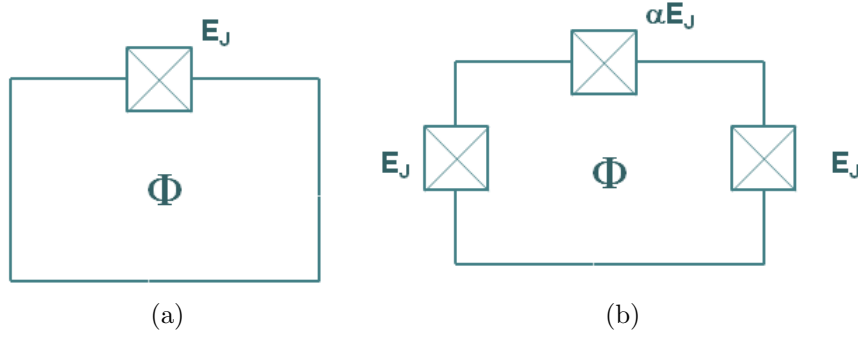


Figure 4.3: Flux Qubit with external flux Φ (a) rf-SQUID (b) Three junction loop as the flux qubit, with a junction that has slightly different capacitance and coupling energy.

by choosing material with large superconducting gap [16], the only relevant degree of freedom is the number of Cooper pairs N on the island, and the system is described by the Hamiltonian [3]

$$H = E_C \sum_N (N - N_g)^2 |N\rangle \langle N| - \frac{E_J}{2} \sum_N |N+1\rangle \langle N| + H.c. \quad (4.2.2)$$

where we have used $\cos \phi = 1/2 (\sum_N |N+1\rangle \langle N| + H.c.)$. When N_g is between $[0, 1]$, the system can be described as two level system with the Hamiltonian

$$H = -\frac{E_c}{2} (1 - 2N_g) \sigma_z - \frac{E_J}{2} \sigma_x \quad (4.2.3)$$

with the charge basis $|0\rangle$ and $|1\rangle$. We noticed that in the lowest energy state $N = 0, 1$, $N_g = 0.5$ will produce degenerate states of superposition of states $|0\rangle$ and $|1\rangle$.

4.2.2 Flux Qubit

Operating in the regime $E_J \gg E_c$, flux qubit is the magnetic analogue of the electrostatic CPB. The simplest form of the flux qubit is called rf-SQUID², which consist of a loop intersect by a Josephson junction. When a magnetic field is applied through the loop, a clockwise or counterclockwise supercurrent is induced, and flux quantization requires that the fluxoid, which includes the Josephson phase (c.f. Eq. (4.1.5)) and total magnetic

²SQUID stands for Superconducting Quantum Interference Device.

field satisfy

$$\frac{\Phi_0}{2\pi}\phi + \Phi_e = m\Phi_0 \quad (4.2.4)$$

where m is an integer. Despite of its simplicity, this form however requires large self inductance (large loops) in order to operate, which makes it susceptible to external noise. Hence an alternative is proposed by Mooji et al. (1999) to use a smaller superconducting loop with three junction, with one of the junction is smaller in area and thus has a smaller critical current to increase the inductance of the loop. The potential of this system is given by [16]

$$U(\phi_1, \phi_2) = -E_J \cos \phi_1 - E_J \cos \phi_2 - \alpha E_J \cos(2\pi \frac{\Phi_x}{\Phi_0} - \phi_1 - \phi_2) \quad (4.2.5)$$

where we have taken into account the fluxoid quantization Eq. (4.2.4). Here ϕ_1 and ϕ_2 is the phase differences across the two junctions respectively. The Josephson energy of the third junction is αE_J . For value of $\alpha > 0.5$, a double well potential is formed with two lowest stable states[16]. The Hamiltonian of the system in the basis of clockwise and counter clockwise circulating current, $|\uparrow\rangle$ and $|\downarrow\rangle$ reads as

$$H = I_p(\Phi_e - \Phi_0/2)\sigma_z - \frac{\Delta}{2}\sigma_x \quad (4.2.6)$$

where I_p is the persistence current of the qubit, and Δ is the tunnelling splitting between the two states. When the externally applied magnetic flux $\Phi_e = \Phi_0/2$, the double well potential becomes symmetrical, and the two eigenfunctions becomes superpositions of the two basis states with the energy difference of Δ .

4.2.3 General Form and Controllability

In order to achieve more controllability on the superconducting qubits, the single Josephson junction may be replaced by two junctions in a loop. This configuration, which is known as dc SQUID is biased by an external flux Φ_e , which can be tune to control the

potential energy:

$$U = -E_J \cos \left(\phi + \pi \frac{\Phi_e}{\Phi_0} \right) - E_J \cos \left(\phi - \pi \frac{\Phi}{\Phi_0} \right) \quad (4.2.7)$$

$$= -2E_J \cos \left(\pi \frac{\Phi_e}{\Phi_0} \right) \cos \phi \quad (4.2.8)$$

$$= -2\tilde{E}_J(\Phi_e) \cos \phi \quad (4.2.9)$$

where $\tilde{E}_J(\Phi_e) = E_J \cos \left(\pi \frac{\Phi_e}{\Phi_0} \right)$.³ Upon applying this to both charge and flux qubits, we obtain a general form of

$$H_1 = -\frac{1}{2} (\epsilon(V_g) \sigma_z + \Delta(\Phi_{e1}) \sigma_x) \quad (4.2.10)$$

with

$$\epsilon(V_g) = E_C(1 - 2N_g) \quad (4.2.11)$$

$$\Delta(\Phi_{e1}) = E_{J1} \cos \left(\pi \frac{\Phi_{e1}}{\Phi_0} \right) \quad (4.2.12)$$

for charge qubit, and

$$H_2 = -\frac{1}{2} (\epsilon(\Phi_{e2}) \sigma_z + \Delta(\Phi_\alpha) \sigma_x) \quad (4.2.13)$$

with

$$\epsilon(\Phi_{e2}) = 2I_p \left(\Phi_{e2} - \frac{1}{2} \Phi_0 \right) \quad (4.2.14)$$

$$\Delta(\Phi_\alpha) = \alpha E_{J2} \cos \left(\pi \frac{\Phi_\alpha}{\Phi_0} \right) \quad (4.2.15)$$

with this we achieved controllability on the various parameters of the system.

³This junction also possess effective junction capacitance of $C_{total} = 2C_J$.

Chapter 5

Dispersive coupling of superconducting qubits

With the dispersive theory and superconducting qubits background at our disposal, we are now ready to discuss the dispersive coupling of superconducting qubit. Here we shall present a hybrid quantum circuit, which consist of charge and flux qubit coupled by a resonator known as nanomechanical resonator. This proposed hybrid circuit provides a promising quantum memory by utilizing long decoherence time of the flux qubit operating at optimal (degeneracy) point.

5.1 Introduction

As described in the previous chapter, charge and flux qubits are two different types of superconducting Josephson-junction (JJ) qubits. Each type of qubits has its own inherent features. For example, the charge qubit can be controlled flexibly through external tunable parameter such as gate voltage and applied magnetic flux. Hence it can serve as operational qubit. As for the flux qubit, it is harder to control the tunnel coupling amplitude but it has a longer decoherence time compared with charge qubit as it is less affected by the charge fluctuation. Thus the flux qubit has the potential of acting as a memory qubit. The hybrid quantum circuit will takes the advantage of both, by having the charge qubit to perform information processing, while having the flux qubit as storage.

5.2 Nanomechanical resonator

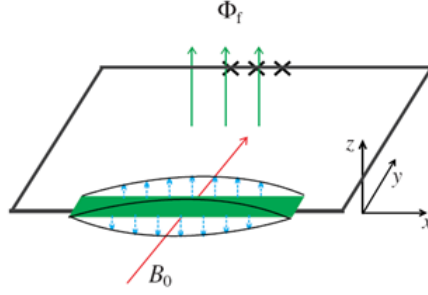


Figure 5.1: Nanomechanical resonator coupled to a 3-junction flux qubit, with B field acting on the transverse direction

The success of circuit QED experiments, such as realization of on chip cavity QED by coupling of charge qubit to coplanar transmission line [3, 15] motivate us to investigate the possibility of substituting other quantum solid-state devices for the resonator. A possible candidates for this solid state device is the nanomechanical resonator (NAMR), which has the properties of high quality factor, low dissipation and small in scale. At attainable temperature of the dilution refrigerator, NAMR is known to approach the quantum regime [13]. NAMR operating at hundreds of MHz to 1GHz has been attained [23], hence is favorable for incorporating it in the superconducting qubit circuit which also operates in the microwave regime.

By considering only the fundamental flexural mode of the NAMR, we can model the NAMR as a harmonic oscillator with a high-Q mode of frequency ω and mass m . The Hamiltonian without dissipation can be expressed as

$$H_f = \frac{p_x^2}{2m} + \frac{1}{2}m\omega^2 x^2 \quad (5.2.1)$$

with the canonical momentum p and coordinate x . Upon canonical quantization, x and p can be represented by the creation and annihilation operator, $a^\dagger$ and a as $x = \frac{1}{\sqrt{2m\omega}}(a + a^\dagger)$, $p = i\frac{\sqrt{m\omega}}{2}(a^\dagger - a)$. Thus the Hamiltonian can be rewritten as

$$H_f = \hbar\omega(a^\dagger a + \frac{1}{2}) \quad (5.2.2)$$

which is the familiar free field Hamiltonian described in Sec. 2.1.

5.2.1 Coupling methods

There are several ways of coupling the nanomechanical to the circuit, for example capacitance coupling and flux coupling. However, in order to achieve controllability in coupling, as well as independence between the free Hamiltonian and the interaction Hamiltonian, we will deploy the method described in [23], which utilize an external transverse B field. In this configuration, the perpendicular arrangement of the magnetic field and the direction of the current generates a Lorentz force in the plane of the beam. The oscillation of the NAMR is modulated by these Lorentz forces, and in this way the quantized harmonic oscillation mode of the beam is coupled to the quantum state of the qubit system.

5.3 Coupling Scheme

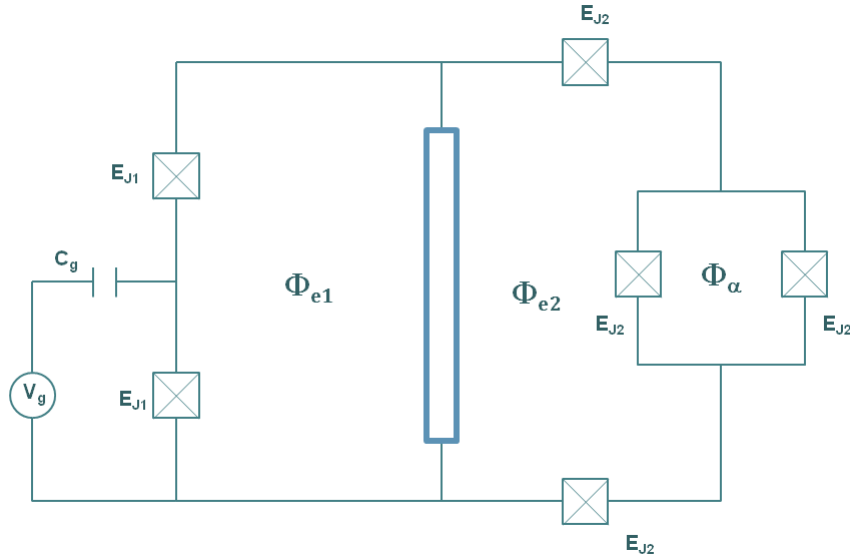


Figure 5.2: Schematic diagram of a hybrid quantum circuit with the charge (left) and flux (right) qubits connected to a NAMR.

The schematic diagram of the hybrid system is shown in Fig. 5.2. The Hamiltonian of the above system takes the form of

$$H = H_1 + H_2 + H_f + H_{\text{int}} \quad (5.3.1)$$

where H_1 , H_2 and H_f are same as defined before (See Sec. 4.2.3). Thus we focus on the interaction Hamiltonian H_{int} , which is given by [23]

$$H_{\text{int}} = B_0(I_1 + I_2)L\delta_z \quad (5.3.2)$$

where B_0 is the transverse magnetic field, L is the effective length of the resonator and $\delta_z = \sqrt{\hbar/2m\omega}$ is the zero-point motion in the z direction. The current I_j in term of the charge/current basis states are

$$I_1 = I_c \sin(\pi \frac{\Phi_{e1}}{\Phi_0}) \sigma_x^1 \quad (5.3.3)$$

$$I_2 = I_p \sigma_z^2 \quad (5.3.4)$$

in which Φ_{e1} is the magnetic flux at the dc SQUID of charge qubit, Φ_0 is the flux quantum $h/2e$ and I_c , I_p are the current on charge and flux qubit respectively. Notice that when $\Phi_e = \Phi_0$, the interaction between the charge qubit and NAMR will be turned off. Hence H_{int} can be rewritten as

$$H_{\text{int}} = \hbar(g_1 \sigma_x^1 + g_2 \sigma_z^2)(a + a^\dagger) \quad (5.3.5)$$

with

$$g_1 = I_c \sin(\pi \frac{\Phi_{e1}}{\Phi_0}) / \hbar \quad (5.3.6)$$

$$g_2 = \frac{I_p}{\hbar}$$

Putting everything together and express in term of the energy eigenbasis (See appendix B.1), the total Hamiltonian read as

$$H = H_0 + H_{\text{int}} \quad (5.3.7)$$

with

$$H_0 = \frac{\hbar}{2}\omega_0^1\tilde{\sigma}_z^1 + \frac{\hbar}{2}\omega_0^2\tilde{\sigma}_z^2 + \hbar\omega(a^\dagger a) \quad (5.3.8)$$

$$H_{\text{int}} = \hbar\left(g_1(\sin\theta_1\tilde{\sigma}_z^1 + \cos\theta_1\tilde{\sigma}_x^1) + g_2(\cos\theta_2\tilde{\sigma}_z^1 - \sin\theta_2\tilde{\sigma}_x^2)\right)(a + a^\dagger) \quad (5.3.9)$$

with

$$\tan\theta_j = \Delta_j/\epsilon_j \quad (5.3.10)$$

$$\omega_0^j = \sqrt{\epsilon_j^2 + \Delta_j^2} \quad (5.3.11)$$

By setting $\Phi_{ej} = \Phi_0/2$, we have $\Delta(\Phi_{e1}) = \epsilon(\Phi_{e2}) = 0$, which implies $\sin\theta_1 = \cos\theta_2 = 0$ and $\cos\theta_1 = \sin\theta_2 = 1$. We further choose $\epsilon(V_g) = \Delta(\Phi_\alpha) = \omega_0$, we obtain $\omega_0^j = \omega_0$. Eventually the Hamiltonian take a simple form of

$$H = \frac{\hbar}{2}\omega_0(\tilde{\sigma}_z^1 + \tilde{\sigma}_z^2) + \hbar\omega(a^\dagger a) \quad (5.3.12)$$

$$+ \hbar(g_1\tilde{\sigma}_x^1 + g_2\tilde{\sigma}_x^2)(a + a^\dagger) \quad (5.3.13)$$

which is the two qubit-resonator Hamiltonian, Eq. (3.3.1) we defined in Sec. 3.3 and the rest follows.

5.4 Remarks on dispersive regime operation

For the sake of illustration, here we give a rough estimate of the interqubit coupling strength as well as the regime of operation. The typical system parameters, which we obtained from [24, 23, 3]. Firstly, the frequencies of the qubit and resonator are given by $\omega = 100\text{MHz}$, $\omega_0 = 5\text{GHz}$. Hence we are in the large detune regime where dispersive theory applies. Since ω is fixed and ω_0 is larger, we are in the case of positive detuning with ω fixed. This corresponds to the case where the interqubit coupling strength is bounded even though we are allowed to further increase the coupling strength.

As for the coupling strength, parameters required are $I_p = 660\text{nA}$, $I_c = 50\text{nA}$, $L_0 = 3.9\mu\text{m}$, $\delta_z = 2.6 \times 10^{-13}\text{m}$.¹ The lifetime of the charge and flux qubit are $0.1 - 1\mu\text{s}$ and $1 - 10\mu\text{s}$. For the NAMR operating at 100MHz , $Q = 2 \times 10^4$ gives $1/\kappa = Q/\omega = 200\mu\text{s}$. When the $B_0 = 5\text{mT}$, we have $g_1 \approx 4.01 \times 2\pi\text{MHz}$, $g_2 \approx 0.31 \times 2\pi\text{MHz}$, which corresponds to $0.016\mu\text{s}$ and $0.21\mu\text{s}$. Although the flux qubit is in strong coupling regime, the charge qubit is in the weak coupling regime. Hence larger coupling strength is required, which can be achieved by simply increasing the magnetic field, for instance by setting $B_0 = 50\text{mT}$. As a result, this hybrid system will operate in the form of strong coupling for charge qubit, and ultrastrong coupling for flux qubit. Therefore RWA is not sufficient to describe the system and we require NRWA description for more accurate estimation on the interqubit coupling strength.

Although RWA does not apply, since ω_0 and ω is finite in our case, we can obtain an approximation of the two-qubit evolution operator by making second order RWA on the NRWA result as shown in Eq. (3.3.11). The resulting unitary operator is then given by

$$U_{\text{NRWA}} \approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos J_{\text{NRWA}}t & i \sin J_{\text{NRWA}}t & 0 \\ 0 & i \sin J_{\text{NRWA}}t & \cos J_{\text{NRWA}}t & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (5.4.1)$$

By setting $J_{\text{NRWA}}t = \pi/2$, we obtained an $i\text{SWAP}$ gate (See appendix B.2). This gate can be used to transfer an arbitrary superposition state of the charge qubit to the flux qubit. For instance, we first initialize the flux qubit to the ground state $|g\rangle$ and turn on the magnetic field to couple the qubit. This gives rise to

$$(\alpha|0\rangle + \beta|1\rangle)|g\rangle \Rightarrow |0\rangle(\alpha|0\rangle + i\beta|1\rangle) \quad (5.4.2)$$

The phase i in the flux qubit can then be removed up to a global phase by applying one

¹Noticed that the persistence current I_p is much larger I_c , since the flux qubit operates in the regime where $E_J \gg E_c$.

qubit rotation after the magnetic field is switched off. The information stored in the flux qubit can be retrieved by reversing the process above.

Conclusion

In this project, we have explored and gained more understanding on the ultrastrong coupling regime. Firstly, we discussed the breakdown of RWA from the point of view of time averaging function, which shows that around $g/\omega \approx 0.1$, RWA fails to give accurate description.

After reviewing some of the consequences of NRWA, we focus on the case where qubit and resonator are highly detuned with each other, which is known as the dispersive regime. Inspired by the work done by Zueco et. al, we further extended the discussion by incorporating the time averaging function and derived a general expression for the dispersive Hamiltonian that accounted for both strong and ultrastrong coupling regime. We also showed that for the inequalities $g \ll |\omega_0 - \omega| \ll \omega_0 + \omega$, the magnitude of $|\omega_0 - \omega|$ and $\omega_0 + \omega$ are subjected to the initial condition of the parameters. As a result, this imposed a limit to the coupling strength g for some cases, and also leads to different behavior of the interqubit coupling strength J . Taking into account of the dispersive inequalities, constraints and also time averaging function, we attempted to work out the so-called Intermediate RWA, which displayed the transition between the RWA to the NRWA as the coupling strength increases.

The next part of the project involves the realization of the dispersive coupling of the qubits. We reviewed the superconducting circuits, in particular the charge and flux qubits and showed that they exhibit 2 level system properties. Lastly, we finalized the project by proposing a hybrid quantum system that takes the advantage of both charge and flux

qubit, coupled with the nanomechanical resonator. Since the resonator and the qubit are large detuned with each other, our setup operates at dispersive regime, hence can be describe by the machinery developed earlier. It was also been estimated that our setup requires NRWA for accurate description due to the fact that large coupling strength is necessary for charge qubit to reach strong coupling regime. As a result, flux qubit which have couples stronger to the resonator will be in the regime of ultrastrong coupling.

Since we considered more or less the ideal case, some other work that can be further done includes the consideration of dephasing and decoherence in the qubit resonator system in more detail. Quantitative analysis can be worked out as well, such as the interqubit coupling strength and the number of possible operations can be done. It is also interesting to check if such NAMR resonator system leads to any enhancement of the lifetime of the system.

Appendix A

Preliminary Background

A.1 Interaction Picture

For a Hamiltonian $H = H_0 + H_{\text{int}}$, we consider the evolution of the state $|\psi'\rangle = e^{iH_1 t/\hbar}|\psi\rangle$ where $[H_0, H_1] = 0$ and both are time independent. The state $|\psi\rangle$ satisfy the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H |\psi\rangle \quad (\text{A.1.1})$$

We now take the time derivative of the state ket $|\psi'\rangle$

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\psi'\rangle &= i\hbar \left(e^{iH_1 t/\hbar} iH_1/\hbar + e^{iH_1 t/\hbar} H \right) |\psi\rangle \\ &= -H_1 |\psi'\rangle + e^{iH_1 t/\hbar} (H_0 + H_{\text{int}}) e^{-iH_1 t/\hbar} |\psi'\rangle \\ &= (H_0 - H_1 + e^{iH_1 t/\hbar} H_{\text{int}} e^{-iH_1 t/\hbar}) |\psi'\rangle \end{aligned} \quad (\text{A.1.2})$$

$$= H' |\psi'\rangle \quad (\text{A.1.3})$$

This is a general form for the time independent unitary transformations. In the case where $H_1 = H_0$, we obtain the *interaction picture*, with

$$|\psi\rangle_I = e^{iH_0 t/\hbar} |\psi\rangle \quad (\text{A.1.4})$$

$$H_{\text{int}}^I = e^{iH_0 t/\hbar} H_{\text{int}} e^{-iH_0 t/\hbar} \quad (\text{A.1.5})$$

It can also be shown that for an observable A that does not contain time explicitly, the evolution in interaction picture is given by

$$\frac{dA^I}{dt} = \frac{1}{i\hbar} [A^I, H_0] \quad (\text{A.1.6})$$

which resembles the Heisenberg equation of motion.

A.2 Pauli matrices

The matrix representation for the operators σ_z , σ_- and σ_+ are

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (\text{A.2.1})$$

We may also identify $\sigma_{\pm} = \frac{1}{2}(\sigma_x \pm i\sigma_y)$, where

$$\sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_- = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (\text{A.2.2})$$

The matrices σ_x , σ_y and σ_z are the Pauli spin matrices in the context of spin 1/2 system.

A.3 Quantization of LC circuit

Before we discuss further on the properties of the Josephson junction, we first visit the simplest form of quantum integrated circuit: the LC oscillator which comprises of an capacitor and inductor (Fig. A.1) Kirchhoff's law can be applies to the circuit, giving

$$I_c + I_L = 0 \quad (\text{A.3.1})$$

$$V_c - V_L = 0 \quad (\text{A.3.2})$$

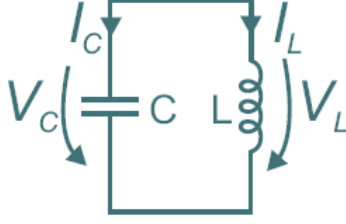


Figure A.1: The LC circuit with branch voltage V and current I , the subscripts L and C correspond to the inductor and capacitance

It is convenient for us to describe the circuit elements in terms of the branch flux Φ and branch charge Q , which are related to the branch voltage and flux by

$$Q(t) = \int dt' I(t') \quad (\text{A.3.3})$$

$$\Phi(t) = \int dt' V(t') \quad (\text{A.3.4})$$

With $Q_c = CV_c$ and $I_L = \Phi_L/L$, we establish

$$C\ddot{\Phi}_L = -\frac{\Phi_L}{L} \quad (\text{A.3.5})$$

This is the analogous to the equation of motion (EOM) for the simple harmonic motion, $m\ddot{x} = -kx$. By identifying $C \rightarrow M$, $\Phi \rightarrow x$ and $L \rightarrow k^{-1}$, the Lagrangian and the Hamiltonian for the LC resonator are

$$L = \frac{1}{2}C\dot{\Phi}_L^2 - \frac{1}{2L}\Phi_L^2 \quad (\text{A.3.6})$$

$$H = \frac{1}{2}\frac{p_L^2}{C} + \frac{1}{2L}\Phi_L^2 \quad (\text{A.3.7})$$

respectively, with

$$p_L = \frac{dL}{d\Phi} = C\dot{\Phi}_L \quad (\text{A.3.8})$$

This Lagrangian description of the EOM provides a starting point for the quantization of LC circuits using *canonical quantization*, in which the canonical variables Φ_L and p will

be changed to quantum operators Φ and P that satisfy the commutator relation

$$[\Phi, P] = i\hbar \quad (\text{A.3.9})$$

In the case of superconducting qubit, electrons comes in pairs called Cooper pairs, hence the classical momentum reads

$$p_L = \frac{dL}{d\Phi} = C\dot{\Phi}_L = CV = Q = 2en \quad (\text{A.3.10})$$

where n is the number of Cooper pairs. We now substitute the classical momentum p with the differential operator

$$P = -i\hbar \frac{\partial}{\partial \Phi} \quad (\text{A.3.11})$$

Or with Eq. (A.3.10) and (4.1.5), we can rewrite this as

$$N = -i \frac{\partial}{\partial \Phi} \quad (\text{A.3.12})$$

where N is the pair number operator. The phase operator ϕ (obtain from Φ) and N has a particular simple commutator relation

$$[\phi, N] = i \quad (\text{A.3.13})$$

with implies that the phase and the number of pairs electron are canonically conjugate.

Appendix B

Qubit Gate Operation

B.1 The two level state

The general 1-qubit Hamiltonian in basis state of σ_z , $|0\rangle$ and $|1\rangle$ has the form[22]

$$H = -\frac{1}{2}(\epsilon\sigma_z + \eta\sigma_x) = -\frac{1}{2}\begin{pmatrix} \epsilon & \eta \\ \eta & -\epsilon \end{pmatrix} \quad (\text{B.1.1})$$

which can be parametrized by rewriting the Hamiltonian as

$$H = \frac{1}{2}\Omega(\theta)(\cos\theta\sigma_z + \sin\theta\sigma_x) \quad (\text{B.1.2})$$

where the energy splitting between the eigenstates is

$$\Omega(\theta) = \sqrt{\epsilon^2 + \eta^2} \quad (\text{B.1.3})$$

with the mixing angle

$$\theta = \tan^{-1} \frac{\eta}{\epsilon} \quad (\text{B.1.4})$$

The eigenstates, which we denote as $|+\rangle$ and $|-\rangle$ are

$$|+\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle, \quad (\text{B.1.5})$$

$$|-\rangle = -\sin \frac{\theta}{2} |0\rangle + \cos \frac{\theta}{2} |1\rangle \quad (\text{B.1.6})$$

which we may use to reexpress Eq. (B.1.3) in term of $\tilde{\sigma}_z = \cos \theta \sigma_z + \sin \theta \sigma_x$ and obtain

$$H = \frac{1}{2} \Omega(\theta) \tilde{\sigma}_z \quad (\text{B.1.7})$$

B.2 One qubit operation

For the Hamiltonian Eq. (B.1.1), if we only turn on η , the quantum states (in $|0\rangle$, $|1\rangle$ basis) will evolve according to the unitary transformation

$$U_x = \exp \frac{-iHt}{\hbar} = \exp \left(\frac{i\eta t}{2\hbar} \sigma_x \right) = \begin{pmatrix} \cos \frac{\phi}{2} & i \sin \frac{\phi}{2} \\ i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix} \quad (\text{B.2.1})$$

where $\phi = \eta t / \hbar$. For $\phi = \pi$ the operation is equivalent to the NOT gate

$$U_x(\phi = \pi) = i \cdot \text{NOT} = i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (\text{B.2.2})$$

which will permutes the vectors $|0\rangle \rightarrow |1\rangle$ and vice versa up to an global phase i . Setting ϕ to $\pi/2$ will give rise to a logic gate which will produce an equal-weight of superposition states between $|0\rangle$ and $|1\rangle$. On the other hand, switching on ϵ will produces the operation which will induce phase shift between $|0\rangle$ and $|1\rangle$:

$$U_x = \exp \left(\frac{i\epsilon t}{2\hbar} \sigma_z \right) = \begin{pmatrix} e^{i\varphi/2} & 0 \\ 0 & e^{-i\varphi/2} \end{pmatrix} \quad (\text{B.2.3})$$

where $\varphi = \epsilon t / \hbar$. Hence with this sequence of σ_z and σ_x rotation any unitary transformation of the qubit state can be performed.

B.3 Two qubit operation

The two qubit operation can be induced by coupling between qubits. We shall illustrate with two examples, with the basis $|0_i, 0_j\rangle, |0_i, 1_j\rangle, |1_i, 0_j\rangle, |1_i, 1_j\rangle$:

1. XY coupling

This coupling corresponds to $J\sigma_-^j\sigma_+^k + \sigma_+^j\sigma_-^k$. Upon diagonalization, we obtain the unitary operator

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \gamma & i \sin \gamma & 0 \\ 0 & i \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{B.3.1})$$

with $\gamma = 2Jt/\hbar$. For $\gamma = \pi/2$, we have the operator known as iSWAP gate,

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{B.3.2})$$

which interchanges the states $|0_i, 1_j\rangle$ with $|1_i, 0_j\rangle$ with an overall phase i , while for $\gamma = \pi/4$ it becomes $i\sqrt{\text{SWAP}}$ gate

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & i\frac{1}{\sqrt{2}} & 0 \\ 0 & i\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{B.3.3})$$

which transform the state $|0_i, 1_j\rangle$ into an entangled state $1/\sqrt{2}(|e_j, e_k\rangle + i|g_j, e_k\rangle)$

2. Ising xx coupling This coupling corresponds to $J\sigma_x^j\sigma_x^k$. Upon diagonalization, the

unitary operator reads

$$U = \begin{pmatrix} \cos \gamma & 0 & 0 & i \sin \gamma \\ 0 & \cos \gamma & i \sin \gamma & 0 \\ 0 & i \sin \gamma & \cos \gamma & 0 \\ i \sin \gamma & 0 & 0 & \cos \gamma \end{pmatrix} \quad (\text{B.3.4})$$

If we rearrange the basis as $|0_i, 0_j\rangle, |1_i, 1_j\rangle, |0_i, 1_j\rangle, |1_i, 0_j\rangle$, it becomes

$$U = \begin{pmatrix} \cos \gamma & i \sin \gamma & 0 & 0 \\ i \sin \gamma & \cos \gamma & 0 & 0 \\ 0 & 0 & \cos \gamma & i \sin \gamma \\ 0 & 0 & i \sin \gamma & \cos \gamma \end{pmatrix} \quad (\text{B.3.5})$$

in which apart of having the properties of XY coupling, the ground state $|0_i, 0_j\rangle$ and the excited state $|1_i, 1_j\rangle$ also exhibit swapping and entanglement in their own subspace.

Appendix C

Dispersive Regime: Derivation

C.1 General expression for one qubit

C.1.1 Basic Commutator Relations

Before the derivation we shall give a list of the basic commutator relations.

- Pauli Matrices

$$[\sigma_j, \sigma_k] = 2i\epsilon_{jkl}\sigma_l \quad (\text{C.1.1})$$

$$\{\sigma_i, \sigma_j\} = 0 \quad (\text{C.1.2})$$

$$\sigma_j^2 = 1 \quad (\text{C.1.3})$$

$$[\sigma_-, \sigma_+] = -\sigma_z \quad (\text{C.1.4})$$

$$[\sigma_z, \sigma_\pm] = \pm 2\sigma_\pm \quad (\text{C.1.5})$$

Other relations are

$$\sigma_\pm \sigma_\mp = \frac{1}{2}[1 \pm \sigma_z] \quad (\text{C.1.6})$$

$$\sigma_z \sigma_\pm = \pm \sigma_\pm \quad (\text{C.1.7})$$

- Creation and Anihilation operators

$$[a, a^\dagger] = 1 \quad (\text{C.1.8})$$

$$\{a, a^\dagger\} = 2a^\dagger a + 1 \quad (\text{C.1.9})$$

$$[a^\dagger a, a] = -a \quad (\text{C.1.10})$$

$$[a^\dagger a, a^\dagger] = a^\dagger \quad (\text{C.1.11})$$

The Hamiltonian for one qubit coupled to the resonator is

$$\begin{aligned} H &= H_0 + H_{\text{int}} \\ &= \frac{\hbar}{2}\omega_0\sigma_z + \hbar\omega a^\dagger a + \hbar g^{(r)}X_+ + \hbar g^{(ar)}Y_+ \end{aligned} \quad (\text{C.1.12})$$

where ω_0 and ω are qubit and resonator frequency respectively. The coupling strength are written in the form of $g^{(r)}$ and $g^{(ar)}$ to keep track of the rotating and counter-rotating terms. $X_\pm$ and $Y_\pm$ are defined as

$$X_\pm = a\sigma_\pm \pm a^\dagger\sigma_\mp, \quad (\text{C.1.13})$$

$$Y_\pm = a\sigma_\mp \pm a^\dagger\sigma_\pm \quad (\text{C.1.14})$$

In order to diagonalize the Hamiltonian by removing the first order term in coupling, we introduce the unitary transformation,

$$U = \exp(\lambda_\Delta X_- + \lambda_\Sigma Y_-) \quad (\text{C.1.15})$$

where $\lambda_\Delta = g^{(r)}/\Delta$, $\lambda_\Sigma = g^{(ar)}/\Sigma$ with $\Delta = \omega_0 - \omega$ and $\Sigma = \omega_0 + \omega$, the Hamiltonian can be transformed up to second order in λ_Δ and λ_Σ :

$$\begin{aligned} H_{\text{disp}} &= U^\dagger H U \\ &= H + \lambda_\Delta [H, X_-] + \lambda_\Sigma [H, Y_-] \\ &\quad + \frac{1}{2} \left(\lambda_\Sigma^2 [[H, X_-], X_-] + \lambda_\Delta \lambda_\Sigma [[H, X_-], Y_-] + \lambda_\Delta^2 [[H, Y_-], Y_-] + \lambda_\Sigma \lambda_\Delta [[H, Y_-], X_-] \right) \end{aligned} \quad (\text{C.1.16})$$

We shall now evaluate the respective commutators. The building blocks are

$$[X_+, X_-] = \sigma_z [2a^\dagger a + 1] + 1 \quad (\text{C.1.17})$$

$$[Y_+, Y_-] = \sigma_z [2a^\dagger a + 1] - 1 \quad (\text{C.1.18})$$

$$[Y_\pm, X_\mp] = \pm \sigma_z (a^2 + (a^\dagger)^2) \quad (\text{C.1.19})$$

$$[H_0, X_-] = -\hbar \Delta X_+ \quad (\text{C.1.20})$$

$$[H_0, Y_-] = -\hbar(\epsilon + \omega) Y_+ \quad (\text{C.1.21})$$

$$[[H_0, X_-], X_-] = -\hbar \Delta [X_+, X_-] \quad (\text{C.1.22})$$

$$[[H_0, X_-], Y_-] = -\hbar \Delta [X_+, Y_-] \quad (\text{C.1.23})$$

$$[[H_0, Y_-], Y_-] = -\hbar \Sigma [Y_+, Y_-] \quad (\text{C.1.24})$$

$$[[H_0, Y_-], X_-] = -\hbar \Sigma [Y_+, X_-] \quad (\text{C.1.25})$$

We first break down the terms into

$$[H, X_-] = [H_0, X_-] + \hbar g^{(r)} [X_+, X_-] + g^{(ar)} [Y_+, X_-] \quad (\text{C.1.26})$$

$$[H, Y_-] = [H_0, Y_-] + \hbar g^{(r)} [Y_+, Y_-] + g^{(ar)} [Y_+, X_-] \quad (\text{C.1.27})$$

Since we only expand up to second order, we have

$$[[H, A], B] \approx [[H_0, A], B] \quad (\text{C.1.28})$$

Putting all together and group the terms, we eventually obtain

$$\begin{aligned}
H_{\text{disp}} = H_0 + \hbar \Bigg\{ & g^{(r)} X_+ + g^{(ar)} Y_+ - \Delta X_+ \frac{g^{(r)}}{\Delta} - (\Sigma Y_+) \frac{g^{(ar)}}{\Sigma} \\
& + (g^{(r)} \lambda_\Delta + g^{(ar)} \lambda_\Sigma) [X_+, X_-] + (g^{(r)} \lambda_\Sigma - g^{(ar)} \lambda_\Delta) [X_+, Y_-] \\
& - \frac{1}{2} (\Delta \lambda_\Delta^2 + \Sigma \lambda_\Sigma^2) [Y_+, Y_-] - \frac{\lambda_\Delta \lambda_\Sigma}{2} (\Delta + \Sigma) [Y_+, X_-] \Bigg\}
\end{aligned} \tag{C.1.29}$$

Hence the first order term will cancel off, and since $[X_+, X_-] = [Y_+, Y_-]$ when constant term dropped (which only lead to overall energy shift), and $[X_+, Y_-] = -[Y_+, X_-]$, it reduces to

$$\begin{aligned}
H_{\text{disp}} = H_0 + \frac{\hbar}{2} \left(\frac{g^{(r)2}}{\Delta} + \frac{g^{(ar)2}}{\Sigma} \right) [X_+, X_-] \\
+ \frac{\hbar}{2} \left(\frac{g^{(r)} g^{(ar)}}{\Delta} + \frac{g^{(ar)} g^{(r)}}{\Sigma} \right) [Y_+, X_-] \\
= H_0 + \frac{\hbar}{2} \left(\frac{g^{(r)2}}{\Delta} + \frac{g^{(ar)2}}{\Sigma} \right) (\sigma_z (2a^\dagger a + 1)) \\
+ \frac{\hbar}{2} \left(\frac{g^{(r)} g^{(ar)}}{\Delta} + \frac{g^{(ar)} g^{(r)}}{\Sigma} \right) (\sigma_z ((a^\dagger)^2 + a^2))
\end{aligned} \tag{C.1.30}$$

giving Eq. (3.2.8).

C.2 General expression for multiple qubit

Following the 1 qubit case, the Hamiltonian takes the form

$$H = \frac{\hbar}{2} \sum_j \omega_0^j \sigma_z^j + \hbar \omega a^\dagger a + \hbar g_j^{(r)} X_+^j + \hbar g_j^{(ar)} Y_+^j \tag{C.2.1}$$

where $X_\pm^j = a \sigma_+^j \pm a^\dagger \sigma_-^j$ and $Y_\pm^j = a \sigma_-^j \pm a^\dagger \sigma_+^j$ [cf. Eq.3.2.3]. The dispersive Hamiltonian can be obtained by the unitary transformation

$$U = \exp \left(\sum_j \lambda_\Delta^j X_-^j + \lambda_\Sigma^j Y_-^j \right) \tag{C.2.2}$$

When it comes to multiple qubit, things get more complicated as we need to consider qubit-qubit interaction as well. Nevertheless it can be simplified by seperating the case of $j = k$ and $j \neq k$

$$\begin{aligned}
H_{N,\text{disp}} &= U^\dagger H U \tag{C.2.3} \\
&= H + \sum_j [\lambda_\Delta^j [H, X_-] + \lambda_\Sigma^j [H, Y_-]] \\
&\quad + \frac{1}{2} \left\{ \left[\lambda_\Delta^j [H, X_-^j] + \lambda_\Sigma^j [H, Y_-^j] \right], \sum_j \lambda_\Delta^j X_-^j + \lambda_\Sigma^j Y_-^j \right\} \\
&= \sum_{j=k} H_{\text{disp}}^j \\
&\quad + \hbar \sum_{j \neq k} \left\{ \lambda_\Delta^j \{ g_k^{(r)} [X_+^k, X_-^j] + g_k^{(ar)} [Y_+^k, X_-^j] \} \right. \\
&\quad + \lambda_\Sigma^j \{ g_k^{(r)} [X_+^k, Y_-^j] + g_k^{(ar)} [Y_+^k, Y_-^j] \} \\
&\quad + \lambda_\Delta^j \lambda_\Delta^k [[H, X_-^j], X_-^k] + \lambda_\Delta^j \lambda_\Sigma^k [[H, X_-^j], Y_-^k] \\
&\quad \left. + \lambda_\Sigma^j \lambda_\Delta^k [[H, Y_-^j], X_-^k] + \lambda_\Sigma^j \lambda_\Sigma^k [[H, Y_-^j], Y_-^k] \right\}
\end{aligned}$$

Together with commutators in the case of one qubit plus

$$[X_+^k, X_-^j] = -[Y_+^k, Y_-^j] = \sigma_-^j \sigma_+^k + \sigma_+^j \sigma_-^k \tag{C.2.4}$$

$$[Y_\pm^j, X_\mp^j] = \sigma_-^j \sigma_-^k + \sigma_+^j \sigma_+^k \tag{C.2.5}$$

$$[H_0, X_-^j] = -\hbar \Delta_j X_+^j \tag{C.2.6}$$

$$[H_0, Y_-^j] = -\hbar \Sigma_j Y_+^j \tag{C.2.7}$$

We eventually obtained

$$\begin{aligned}
H_{\text{N,disp}} &= U^\dagger H U = \sum_j H_{\text{disp}}^j \\
&+ \sum_{j>k} \frac{\hbar}{2} \left\{ \left[g_j^{(r)} g_k^{(r)} \left(\frac{1}{\Delta_j} + \frac{1}{\Delta_k} \right) - g_j^{(ar)} g_k^{(ar)} \left(\frac{1}{\Sigma_j} + \frac{1}{\Sigma_k} \right) \right] (\sigma_-^j \sigma_+^k + \sigma_+^j \sigma_-^k) \right. \\
&\left. + \left[g_j^{(r)} g_k^{(ar)} \left(\frac{1}{\Delta_j} - \frac{1}{\Sigma_k} \right) + g_j^{(ar)} g_k^{(r)} \left(\frac{1}{\Delta_k} - \frac{1}{\Sigma_j} \right) \right] (\sigma_-^j \sigma_-^k + \sigma_+^j \sigma_+^k) \right\}
\end{aligned} \tag{C.2.8}$$

as Eq. (3.3.4).

C.3 Plotting of curves under constraints

For the frequency shift in one qubit scenario, we have

$$\omega_{\text{RWA}} = \omega \pm g^2/\Delta, \tag{C.3.1}$$

$$\omega_{\text{NRWA}} = \omega \pm g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \tag{C.3.2}$$

for RWA and NRWA. Under the constraints discussed in Sec. 3.1.2, there are 4 different cases to be considered here

C.3.1 $\Delta = \omega_0 - \omega > 0$

Increase ω_0 , keeping ω constant

In this case, set of equations are

$$\Delta = \alpha g, \tag{C.3.3}$$

$$\Sigma = 2\omega + \alpha g \tag{C.3.4}$$

Using these set of equations in C.3.1, we obtained

$$\frac{\omega_{\text{RWA}}}{\omega} = \left(\frac{g}{\omega}\right)^2 \left[\frac{1}{\alpha \frac{g}{\omega}} \right], \quad (\text{C.3.5})$$

$$\frac{\omega_{\text{NRWA}}}{\omega} = \left(\frac{g}{\omega}\right)^2 \left[\frac{1}{\alpha \frac{g}{\omega}} + \frac{1}{2 + \alpha \frac{g}{\omega}} \right] \quad (\text{C.3.6})$$

Decrease ω , keeping ω_0 constant

Now the set of equations becomes

$$\Delta = \alpha g, \quad (\text{C.3.7})$$

$$\Sigma = 2\omega_0 - \alpha g \quad (\text{C.3.8})$$

With these set of equation and Eq. (C.3.1), we establish

$$\frac{\omega_{\text{RWA}}}{\omega_0} = \left(\frac{g}{\omega_0}\right)^2 \left[\frac{1}{\alpha \frac{g}{\omega_0}} \right], \quad (\text{C.3.9})$$

$$\frac{\omega_{\text{NRWA}}}{\omega_0} = \left(\frac{g}{\omega_0}\right)^2 \left[\frac{1}{\alpha \frac{g}{\omega_0}} + \frac{1}{2 - \alpha \frac{g}{\omega_0}} \right] \quad (\text{C.3.10})$$

C.3.2 $\Delta = \omega_0 - \omega < 0$

We have more or less similar situation here but the sign on Δ will be different. To avoid conflict we set $-\Delta = \gamma g$

Increase ω , keeping ω_0 constant

In this case, set of equations are

$$\Delta = -\gamma g, \quad (\text{C.3.11})$$

$$\Sigma = 2\omega_0 + \gamma g \quad (\text{C.3.12})$$

Using these set of equations in C.3.1, we obtained

$$\frac{\omega_{\text{RWA}}}{\omega_0} = \left(\frac{g}{\omega_0}\right)^2 \left[-\frac{1}{\gamma \frac{g}{\omega_0}}\right], \quad (\text{C.3.13})$$

$$\frac{\omega_{\text{NRWA}}}{\omega_0} = \left(\frac{g}{\omega_0}\right)^2 \left[-\frac{1}{\gamma \frac{g}{\omega_0}} + \frac{1}{2 + \gamma \frac{g}{\omega_0}}\right] \quad (\text{C.3.14})$$

Decrease ω_0 , keeping ω constant

Now the set of equations becomes

$$\Delta = \gamma g, \quad (\text{C.3.15})$$

$$\Sigma = 2\omega - \gamma g \quad (\text{C.3.16})$$

With these set of equation and Eq. (C.3.1), we establish

$$\frac{\omega_{\text{RWA}}}{\omega} = \left(\frac{g}{\omega}\right)^2 \left[-\frac{1}{\gamma \frac{g}{\omega}}\right], \quad (\text{C.3.17})$$

$$\frac{\omega_{\text{NRWA}}}{\omega} = \left(\frac{g}{\omega}\right)^2 \left[-\frac{1}{\gamma \frac{g}{\omega}} + \frac{1}{2 - \gamma \frac{g}{\omega}}\right] \quad (\text{C.3.18})$$

For the two qubit case, the interqubit coupling strength J has similar expression,

$$J_{\text{RWA}} = \omega \pm g^2/\Delta, \quad (\text{C.3.19})$$

$$J_{\text{NRWA}} = \omega \pm g^2\left(\frac{1}{\Delta} - \frac{1}{\Sigma}\right) \quad (\text{C.3.20})$$

when the qubit's energy are degenerate. Hence the whole analysis above can be repeated similarly for $J_{\text{RWA,NRWA}}$.

Bibliography

- [1] J. Bardeen, L.N. Cooper, and J.R. Schrieffer. Theory of superconductivity. *Physical Review*, 108(5):1175, 1957.
- [2] A. Blais, J. Gambetta, A. Wallraff, DI Schuster, SM Girvin, MH Devoret, and RJ Schoelkopf. Quantum-information processing with circuit quantum electrodynamics. *Physical Review A*, 75(3):032329, 2007.
- [3] A. Blais, R.S. Huang, A. Wallraff, SM Girvin, and RJ Schoelkopf. Cavity quantum electrodynamics for superconducting electrical circuits: An architecture for quantum computation. *Physical Review A*, 69(6):062320, 2004.
- [4] D. Bruss and G. Leuchs. *Lectures on quantum information*. Wiley, 2007.
- [5] Raymond Chiao and John Garrison. *Quantum Optics (Oxford Graduate Texts)*. Oxford University Press, USA, August 2008.
- [6] J. Clarke and F.K. Wilhelm. Superconducting quantum bits. *Nature*, 453(7198):1031–1042, 2008.
- [7] MH Devoret, A. Wallraff, and JM Martinis. Superconducting qubits: A short review. *Arxiv preprint cond-mat/0411174*, 2004.
- [8] P. Forn-Díaz, J. Lisenfeld, D. Marcos, J. J. García-Ripoll, E. Solano, C. J. P. M. Harman, and J. E. Mooij. Observation of the Bloch-Siegert Shift in a Qubit-Oscillator System in the Ultrastrong Coupling Regime. May 2010.
- [9] Mark Fox. *Quantum Optics: An Introduction (Oxford Master Series in Physics, 6)*. Oxford University Press, USA, June 2006.

- [10] S. Gleyzes, S. Kuhr, C. Guerlin, J. Bernu, S. Deléglise, U.B. Hoff, M. Brune, J.M. Raimond, and S. Haroche. Quantum jumps of light recording the birth and death of a photon in a cavity. *Nature*, 446(7133):297–300, 2007.
- [11] EK Irish. Generalized rotating-wave approximation for arbitrarily large coupling. *Physical review letters*, 99(17):173601, 2007.
- [12] TD Ladd, F. Jelezko, R. Laflamme, Y. Nakamura, C. Monroe, and JL OBrien. Quantum computers. *Nature*, 464(7285):45–53, 2010.
- [13] MD LaHaye, O. Buu, B. Camarota, and K.C. Schwab. Approaching the quantum limit of a nanomechanical resonator. *Science*, 304(5667):74, 2004.
- [14] I. Lizuain, J. Casanova, J.J. García-Ripoll, JG Muga, and E. Solano. Zeno physics in ultrastrong-coupling circuit QED. *Physical Review A*, 81(6):062131, 2010.
- [15] J. Majer, JM Chow, JM Gambetta, J. Koch, BR Johnson, JA Schreier, L. Frunzio, DI Schuster, AA Houck, A. Wallraff, et al. Coupling superconducting qubits via a cavity bus. *Nature*, 449(7161):443–447, 2007.
- [16] Y. Makhlin, G. Schön, and A. Shnirman. Quantum-state engineering with Josephson-junction devices. *Reviews of modern physics*, 73(2):357, 2001.
- [17] T. Niemczyk, F. Deppe, H. Huebl, EP Menzel, F. Hocke, MJ Schwarz, JJ Garcia-Ripoll, D. Zueco, T. Hümmer, E. Solano, et al. Circuit quantum electrodynamics in the ultrastrong-coupling regime. *Nature Physics*, 2010.
- [18] G.M. Reuther, D. Zueco, F. Deppe, E. Hoffmann, E.P. Menzel, T. Weißl, M. Mariani, S. Kohler, A. Marx, E. Solano, et al. Two-resonator circuit QED: Dissipative Theory. *Arxiv preprint arXiv:0911.2657*, 2009.
- [19] RJ Schoelkopf and SM Girvin. Wiring up quantum systems. *Nature*, 451(7179):664–670, 2008.
- [20] Bruce W. Shore and Peter L. Knight. The Jaynes-Cummings Model. *Journal of Modern Optics*, 40(7):1195–1238, 1993.

- [21] S. Stenholm. Foundations of laser spectroscopy. 1984.
- [22] G. Wendin and VS Shumeiko. Superconducting quantum circuits, qubits and computing. *Arxiv preprint cond-mat/0508729*, 2005.
- [23] F. Xue, YD Wang, CP Sun, H. Okamoto, H. Yamaguchi, and K. Semba. Controllable coupling between flux qubit and nanomechanical resonator by magnetic field. *New Journal of Physics*, 9:35, 2007.
- [24] JQ You and F. Nori. Superconducting circuits and quantum information. *Arxiv preprint quant-ph/0601121*, 2006.
- [25] D. Zueco, G.M. Reuther, S. Kohler, and P. Hänggi. Qubit-oscillator dynamics in the dispersive regime: Analytical theory beyond the rotating-wave approximation. *Physical Review A*, 80(3):033846, 2009.